Finding the edges is not enough: an evaluation of edge-function recovery in sparse additive-noise DAG learners

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Abstract

Nonlinear additive-noise DAG learners are valued because every edge is an inspectable function, yet benchmarks score only whether the edges are found. We evaluate the recovered edge functions against ground truth in simulation, using an edge-function error (NRMSE), an interventional-effect error and calibration measures. A screened, group-lasso spline learner (SPADE-scaled) recovers structure almost perfectly on internally standardised data (AUROC 0.997–0.999 for $d = 100$ to 500), but its edge curves keep a median of about 44% of the true amplitude in the main cell and 14–46% across cells (edge-function NRMSE 0.61–0.80) and the interventional-effect curves derived from them are wrong even for direct parents (NRMSE 0.58–0.76). On fresh seeds (75 datasets, eight cells, $d = 100$ and 200) we compare three repairs: an unpenalised refit on the selected support (NRMSE 0.20–0.47), the standard adaptive group lasso (0.24–0.52, structure unchanged, AUROC 0.998), and BASIL-W, a Bayesian model-averaged estimator warm-started from the learner's causal order (0.15–0.29, lower than SPADE-scaled in all 75 datasets). The adaptive group lasso closes about 85% of the gap between SPADE-scaled and BASIL-W at $d = 100$ (about 75% at $d = 500$, two datasets) at no structural cost. BASIL-W is clearly better than the refit only under heavy-tailed noise (by 0.03–0.04 NRMSE under uniform and heteroscedastic noise on 20 independent datasets per cell, where the refit wins 4/20 and 6/20 datasets); it supplies calibrated edge probabilities but costs 0.006–0.032 AUROC, and its 90% credible bands cover only 74–81% of the truth in simulation. BASIL-W is not the best method overall: against external learners at $d \leq 20$, our own Python reimplementation of CAM has lower or equal edge-function error (Laplace noise, $d = 10$: 0.151 against 0.281; pooled over 30 non-independent datasets 0.161 against 0.212), while untuned neural learners are far behind; the benchmark is additive and $d > 20$ was not tested. On a semi-synthetic benchmark built from Sachs the gain of BASIL-W is smaller at $n = 500$ and the simulation story is only partly reproduced; on the real Sachs data no repair improves structure and the refit and BASIL-W are much less stable than SPADE-scaled. None of the repair ingredients is new; our contribution is the diagnosis of amplitude shrinkage in sparse spline DAG learners, the metrics, the evidence that standard fixes remove most of it, the scale at which this was checked, and the negative results.

Keywords. causal discovery; additive noise models; edge functions; group lasso; Bayesian order MCMC; evaluation.

1 Introduction

Differentiable learners for nonlinear additive-noise structural equation models (ANMs) are often motivated by interpretability: each edge $c \to e$ is a univariate function f_{ce} that a practitioner can plot, and the shape (saturating, threshold-like, non-monotone) is meant to carry information beyond the presence of the edge (Agarwal et al., 2021; Liu et al., 2025; Dang et al., n.d.). Published evaluations of such learners, and of nonlinear DAG learners in general, report structure metrics (SHD, SID, AUROC) (Zheng et al., 2020; Lachapelle et al., 2020; Rolland et al., 2022; Montagna et al., 2023c). Whether the returned functions are *correct* is rarely scored, although

a sparsity-inducing group penalty is known to bias the amplitude of the selected components towards zero (Huang et al., 2010; Belloni and Chernozhukov, 2013).

This paper asks a narrow question in a controlled setting: when a sparse spline additive DAG learner finds the edges, how accurate are the functions it returns, how much does that matter for the interventional effects a user would read off the model, and which fixes work? We answer it by simulation with ground-truth edge functions at $d = 100$ to 500, by a pre-registered paired comparison of repair strategies on fresh seeds, by a comparison with published learners at small d, and by a real-data check on the Sachs protein-signalling data where function ground truth does not exist.

Framing. This is an *evaluation study.* It does not propose a new estimator. One of the estimators we compare (BASIL-W) was developed by us in the course of the project; we report it as one candidate among several and disclose that its ingredients are established (Section 2.1). We do not claim that it is the best available learner: at $d \leq 20$ an additive spline baseline from the literature matches or beats it (Section 6.10).

Contributions. We claim the following, and nothing beyond it.

1. **A measurement.** A structure-accurate learner (SPADE-scaled, AUROC 0.997–0.999) returns edge functions with a median of 14–46% of the true amplitude, depending on the cell, and the shortfall persists across four noise families, three sample sizes, a second mechanism family and d up to 500 (Section 6.1). To our knowledge this has not been quantified against ground truth for additive spline or KAN-type DAG learners; the evidence for "not done before" is absence of search hits, not proof.
2. **Metrics.** An edge-function NRMSE over true edges (a missed edge counts as an error of 1), a calibration slope, and an interventional (do-effect) NRMSE for direct and indirect pairs propagated through the learned model, together with edge-probability calibration (ECE) and band coverage (Section 3). The do-effect error follows the established idea of interventional evaluation (Gentzel et al., 2019); we do not claim it as new.
3. **A paired comparison of repairs** that includes the standard adaptive group lasso (Huang et al., 2010), an unpenalised refit, and a Bayesian estimator, with the protocol fixed by git commits before the runs (Section 5). The main finding is deflationary: the adaptive group lasso removes most of the shrinkage without changing the selected structure, and BASIL-W is not shown to be the best method (an independent-DAG replication, a semi-synthetic Sachs benchmark and a comparison with external learners are in Sections 6.5–6.10).
4. **Negative results.** A group-MCP learner with an independence term (SPIRE) loses structure accuracy; the cold-start Bayesian scaling variant (BASIL-S) does not converge at $d \geq 100$; full BASIL does not scale; the pre-registered BASIL-W had a numerical instability in interventional effects; BASIL-W lowers AUROC in every dataset; its credible bands under-cover; and on Sachs the repairs are less stable than the unrepaired learner (Sections 6.8, 6.11, 6.12).

What is not new. The unpenalised refit on a selected support is post-lasso least squares (Belloni and Chernozhukov, 2013). The adaptive group lasso for spline additive models is due to Huang et al. (2010). The ingredients of BASIL-W (closed-form marginal likelihoods with a g-prior, order MCMC, spline-based Bayesian DAGs, model averaging over parent sets) are due to Friedman and Koller (2003), Ni et al. (2015), Tóth et al. (2025) and others. Independence-based criteria combined with adaptive group sparsity appear in a concurrent ICML 2026 paper (Yao et al., 2026), which we know only from its abstract. Interventional evaluation is advocated by Gentzel et al. (2019). We found no prior work that combines these in the way BASIL-W does or that defines the edge-function NRMSE, but this rests on a limited search (Section 2).

Scope of causal claims. Everything here assumes an additive-noise model with causal sufficiency. We make no claim about causal validity of any learned graph or function beyond these assumptions; the heteroscedastic cell violates additivity on purpose and the real-data results are weak.

Outline. Section 2 reviews related work, Section 3 defines the problem and metrics, Section 4 the methods, Section 5 the protocol, Section 6 the results (including the external-baseline comparison in Section 6.10), Section 7 discusses limitations, and Section 8 gives reproducibility details.

2 Related work

Identifiability and continuous DAG learning. Nonlinear additive-noise models make the DAG identifiable (Hoyer et al., 2008; Peters et al., 2014). NOTEARS (Zheng et al., 2018) turned structure search into continuous optimisation with a smooth acyclicity penalty; NOTEARS-MLP (Zheng et al., 2020), GraN-DAG (Lachapelle et al., 2020), GOLEM (Ng et al., 2020) and DAGMA (Bello et al., 2022) extended it to nonparametric models and better-conditioned constraints, and truncated matrix power iteration (Zhang et al., 2022) reduces its cost. Order-based methods split the problem: CAM (Bühlmann et al., 2014) searches orders under additive models and prunes with sparse regression, SCORE (Rolland et al., 2022), NoGAM (Montagna et al., 2023c) and DAS (Montagna et al., 2023b) obtain the order from score matching, and DiffAN does so with diffusion models (Sanchez et al., 2023). These works report structure metrics. Evaluation caveats are well documented: varsortability and R^2-sortability (Reisach et al., 2021, 2023), scale sensitivity of NOTEARS (Kaiser and Sipos, 2021; Seng et al., 2024), and internally standardised generators (Ormaniec et al., 2025), which we use. Montagna et al. (2023a) study robustness of score matching to assumption violations, in the spirit of our heteroscedastic cell.

Additive and spline models, and sparsity bias. Sparse additive models (Ravikumar et al., 2009) and the adaptive group lasso for spline additive models (Huang et al., 2010) select components by group penalties; the latter shows that adaptive weights restore selection consistency by reducing the bias on strong groups. Refitting after selection is classical: least squares after selection has smaller bias than the lasso (Belloni and Chernozhukov, 2013), and the adaptive lasso (Zou, 2006) was designed for the same purpose. In DAG learning, an adaptive lasso has been proposed to avoid hard thresholding that deletes small true edges (Xu et al., 2022). Neural Granger causality uses first-layer group lasso for selection (Tank et al., 2022). A concurrent ICML 2026 paper combines an independence criterion with an adaptive group lasso on first-layer MLP weights to prune constant edges under non-Gaussian, heteroscedastic noise (Yao et al., 2026); we read only its abstract, which states a focus on identifiability and structure, not on recovery of edge amplitudes. We found no study that measures how group-lasso shrinkage distorts the amplitude of recovered edge functions in additive DAG learners; accordingly we treat the plain refit and adaptive penalties as baselines.

Interpretable edge functions. Neural additive models (Agarwal et al., 2021) and KANs (Liu et al., 2025), with critiques by Yu et al. (2024) and Hou et al. (2024), promise readable univariate components. Dagma-DCE (Waxman et al., 2024) replaces weight norms by a derivative-based causal-effect strength. In time series, NAVAR (Bussmann et al., 2021) is additive, and Kuskova et al. (2026) estimate function-valued causal influence from trained NAVAR models and show that equal scalar scores can hide different shapes; in the part we read (abstract only) they do not report an accuracy metric against known functions. Concurvity can make additive components non-identifiable (Siems et al., 2023). To our knowledge no benchmark scores recovered edge functions against ground truth in cross-sectional nonlinear causal discovery. Interventional

evaluation of causal models has been advocated (Gentzel et al., 2019), and recent Bayesian work reports errors of average causal effects (Tóth et al., 2025); our do-effect error follows this line.

Bayesian structure learning. Order MCMC (Friedman and Koller, 2003) samples node orders and sums over parent sets; its prior bias towards graphs compatible with many orders is analysed by Viinikka et al. (2020). Nonlinear Bayesian structure learners include Gaussian process networks with marginal-likelihood scores (Friedman and Nachman, 2000), spline-based Bayesian DAGs with edge and functional selection (Ni et al., 2015), DiBS (Lorch et al., 2021), BayesDAG (Annadani et al., 2023) and structural marginalisation with autoregressive orders for Gaussian-process additive models (Tóth et al., 2025). Wang et al. (2025) build confidence sets for causal orderings under additive errors. Our estimator BASIL-W combines a g-prior closed-form marginal likelihood on spline features with order MCMC warm-started from a continuous learner. Its ingredients are established and we do not claim the inference idea as new.

2.1 What is and is not new here

Not new: the refit, the adaptive group lasso, the Bayesian ingredients, interventional evaluation, the idea of independence plus adaptive group sparsity. Possibly new, subject to the limits of our search: the edge-function NRMSE and the study of amplitude shrinkage in spline additive DAG learners against ground truth; the particular combination BASIL-W and its warm start from a continuous learner. The literature check was performed with web search only, mostly at the level of abstracts and snippets, and absence of a hit is weak evidence.

3 Problem and metrics

Model. Data $x \in \mathbb{R}^d$ follow an additive-noise SEM $x_e = \sum_{c \in \mathrm{pa}(e)} f_{ce}(x_c) + \varepsilon_e$ with ε_e independent of $x_{\mathrm{pa}(e)}$ and a DAG A. Generation is internally standardised (iSCM, Ormaniec et al. 2025): each variable is standardised right after it is generated, so that no learner can exploit marginal variance as a causal-order cue (Reisach et al., 2021). The true edge function in standardised units is $g^*_{ce}(x) = f_{ce}(x)/s_e$, where s_e is the standard deviation of the generated (unstandardised) variable. We always compare centred curves on the observed values of the cause.

Edge-function NRMSE (primary metric). For a true edge $c \to e$ with estimated curve $\hat{g}_{ce}$ (zero if the edge is not selected),

$$\mathrm{NRMSE}_{ce} = \frac{\|\hat{g}_{ce} - g^*_{ce}\|_2}{\|g^*_{ce}\|_2},$$

evaluated on the n observed values of x_c after centring both curves. A missed edge has $\hat{g} = 0$ and an error of 1; an estimate with the right shape and half the amplitude has an error of 0.5. The dataset-level value is the mean over its true edges; the dataset is the unit of analysis. We also report the calibration slope $\langle \hat{g}, g^* \rangle / \langle g^*, g^* \rangle$ (median over edges; 1 is ideal, a slope of 0.44 means 44% of the true amplitude), and the curve R^2.

Do-effect NRMSE. For a pair (c, e) the interventional curve $\tau_{ce}(v) = \mathbb{E}[x_e \mid \mathrm{do}(x_c = v)]$ is evaluated on a 15-point grid between the 5th and 95th percentile of x_c. The truth is obtained by Monte Carlo (1000 draws) from the true SEM. The learned curve is obtained by propagating the intervention through the learned SEM in a topological order of the learned graph (source peeling on edge strength), keeping the observed residuals; 400 rows are used. The error is $\|\hat{\tau} - \tau\|/\|\tau\|$ after centring. Up to 20 *direct* (true parent) and 20 *indirect* (true ancestor, not parent) pairs are drawn per dataset; pairs whose true curve has standard deviation below 0.05 are skipped (about 3.8 of about 40 per dataset, identically for all methods). Indirect pairs test whether errors accumulate along paths.

Structure metrics. Directed AUROC over all ordered pairs against the true adjacency, AUPRC, and orientation accuracy on the true edges (1 if the score for the true direction exceeds the reversed one, 0.5 for ties). Edge scores are the standard deviation of the edge's contribution (learners) or posterior edge probabilities (Bayesian estimators).

Calibration. For the Bayesian estimators: expected calibration error (ECE, 10 bins) of edge probabilities over off-diagonal pairs; pointwise coverage of the 90% credible band of g_{ce} over true edges and rows, where a missed edge has a band concentrated at zero and therefore usually fails to cover.

4 Methods

4.1 SPADE-scaled (SSC)

SPADE (Dang et al., n.d.) represents every edge as a cubic B-spline function on a fixed grid (grid size 8 on $[-3, 3]$, 11 coefficients per edge), regresses each variable on spline functions of its candidate parents, and penalises each edge's coefficient group with a group lasso under an acyclicity constraint. The scaled variant used here (SSC) adds two changes that were fixed on an earlier generator: tree-ensemble screening to $K = 15$ candidate parents per node, and a penalty $\lambda = 0.2/d$ instead of a fixed λ, because a group penalty summed over $d(d-1)$ candidate edges over-penalises as d grows. It is trained for 400 epochs. The edge score is the standard deviation of the edge's contribution on the data. SSC is the baseline against which all repairs are measured.

4.2 Refit (SSR)

The $\lceil 1.5\, d \rceil$ highest-scoring edges of SSC define a support ("top $1.5d$"; the multiplier $k = 1.5$ was chosen on three validation datasets at $d = 100$, among $k \in \{1, 1.5, 2\}$ and uses the knowledge that the generating graphs have about d edges). The model is refit from scratch on this mask with $\lambda = 0$. This is post-lasso least squares in the sense of Belloni and Chernozhukov (2013).

4.3 Adaptive group lasso (SSA)

A second fit of SSC, on the same screened candidates and the same $\lambda = 0.2/d$, with edge-wise weights $w_{ce} = (\|\theta_{ce}\| + 0.01)^{-1}$ computed from the first SSC fit and normalised to mean 1 over the screened edges. This is the adaptive group lasso of Huang et al. (2010) with exponent $\gamma = 1$ and offset $\epsilon = 0.01$. Neither was tuned on any dataset in this paper's main grids. (A sensitivity analysis is in Section 6.6.)

4.4 BASIL-W

BASIL-W (Bayesian spline additive structure learning, warm-started) averages over parent sets and orders. Spline features of each candidate parent are whitened, and for a node i with parent set S ($|S| \le k_{\max} = 3$) the marginal likelihood is available in closed form under a Zellner g-prior with $g = n$ and a per-parent log-odds prior $\log(\pi/(1-\pi))$, $\pi = 2/(d-1)$. The candidates of each node are its top 20 parents ranked by SSC importance. Orders are sampled by Metropolis–Hastings (swap and shift moves) with the node-wise parent-set sum evaluated exactly given the order; two chains of $200d$ iterations (30% burn-in) start at the order obtained by source peeling on the SSC importance matrix, and the chain with the higher end score is kept. Output: posterior edge probabilities; model-averaged curves $\hat{g}_{ce}$ (posterior mean of the $g/(1+g)$-shrunk least-squares coefficients over the 60 highest-weight parent sets); 5–95% pointwise bands from Gaussian posterior draws; and a coefficient tensor for do-effects in which the map from whitened to spline coefficients is ridge-stabilised ($\lambda = 10^{-3}$ times the largest basis eigenvalue of the

Table 1: Pre-registered plans and their commits. All under `research/`.

Plan	Commit	Content
`03_formulation.md`	5e971a5	SPIRE (group-MCP, independence term), $d = 10$
`04_bayes_formulation.md`	6f95e52	BASIL, $d \leq 50$
`05_bayes_scale.md`	045b134	BASIL-S scaling
`06_focus_edge_functions.md`	d5ee082	edge functions at scale, BASIL-W vs refit
`07_do_effect_robustness.md`	f8aa35f	do-effects and robustness grid
`ablation/PLAN.md`	b4626f8	BASIL-W ablations, band calibration
`replication/PLAN.md`	2164313	confirmatory replication, fresh seeds
`adaptive/PLAN.md`	91c240a	adaptive group lasso arm
`real_data/PLAN.md`	6383e6e	Sachs evaluation
`replication2/PLAN.md`	58c4b1f	independent DAGs, 20 per cell, four arms
`adaptive2/PLAN.md`	e3f6316	adaptive arm at $d = 500$, (γ, ϵ) sensitivity
`semisyn/PLAN.md`	03f1942	semi-synthetic benchmark from Sachs
`external/PLAN.md`	c245cc1	external learners at $d \leq 20$

variable; Section 6.12). In all tables "BASIL-W" is this final, stabilised version unless stated. Edge-function NRMSE and band coverage use the model-averaged curves of the unstabilised map (identical to the originally pre-registered curves); the stabilised map affects do-effects only.

4.5 Other estimators studied

Full BASIL (order MCMC over all parents, no screening, feasible only up to $d \approx 50$), BASIL-S (screened candidates, cold-start chains), SPIRE (group-MCP penalty plus an HSIC residual-independence term) and GAM-ridge (a penalised additive regression without DAG constraint) appear in the negative results and on Sachs.

5 Experimental protocol

Pre-registration by commit. Every confirmatory comparison has a plan with hypotheses, thresholds and kill criteria, committed to the git repository before the corresponding runs (Table 1). Deviations are recorded in the result READMEs and in Appendix B. The commit timestamps are those of the repository history; they are evidence of ordering, not an external registry.

Data generator. Erdős–Rényi DAGs in a random order with edge probability $2/(d-1)$ (about d edges, e.g. 83 at $d = 100$), internally standardised (iSCM), $n \in \{500, 1000, 2000\}$. Edge functions are drawn from a mechanism family: family 1 (m1) $\{0.8\tanh(1.5x),\ 0.7\sin(1.5x),\ 0.3(x^2 - 1),\ 0.8x/(1+|x|)\}$; family 2 (m2) $\{0.9x,\ \tanh(2x),\ 0.6(x^2-1),\ \text{logistic}(6(x-0.3))-0.5,\ 0.9\sin(2x)\}$. Noise: uniform$(-0.7, 0.7)$, Gaussian, Laplace (the latter two with matched standard deviation), or heteroscedastic (Gaussian noise scaled by $0.4 + 1.2|x_{\text{pa}_1}|$, which violates the additive-noise assumption).

Cells and seeds. Seeds partition into validation (used for tuning and diagnosis), earlier tests and fresh confirmatory seeds. Validation: seeds 0–9 (tuning of the early SPIRE and BASIL settings and, for the refit multiplier, seeds 0–2 at $d = 100$) and 3000–3004 (band constant κ). Earlier tests: 1000–1004 (and 1000–1029 for $d = 10$ SPIRE), on which the numerical instability of the original BASIL-W do-effects was diagnosed. Confirmatory replication: seeds 2000–2009 in cells C1–C7 and 2000–2004 in C8 (Table 2); no parameter was retuned. Ablations: test seeds

Table 2: Confirmatory replication (seeds 2000–2009; C8: 2000–2004). Generated by `make_tables.py` from `research/replication/results/all.csv`.

Cell	Setting	N	SSC	refit	BASIL-W	ratio	BASIL-W − refit [95% CI]	wins	Holm p (refit)	Holm p (SSC)
C1	d100 uniform	10	0.612	0.213	0.169	0.28	-0.043 [-0.071, -0.016]	8/10	0.082	0.014
C2	d100 Laplace	10	0.803	0.470	0.252	0.31	-0.218 [-0.249, -0.189]	10/10	0.014	0.014
C3	d100 hetero	10	0.768	0.359	0.294	0.38	-0.065 [-0.087, -0.045]	10/10	0.014	0.014
C4	d100 Gaussian	10	0.692	0.278	0.150	0.22	-0.128 [-0.156, -0.100]	10/10	0.014	0.014
C5	d100 n=500	10	0.617	0.231	0.201	0.33	-0.030 [-0.056, -0.004]	6/10	0.131	0.014
C6	d100 n=2000	10	0.611	0.199	0.152	0.25	-0.047 [-0.078, -0.017]	8/10	0.082	0.014
C7	d100 mech. 2	10	0.625	0.270	0.155	0.25	-0.115 [-0.142, -0.088]	10/10	0.014	0.014
C8	d200 uniform	5	0.612	0.203	0.173	0.28	-0.031 [-0.049, -0.011]	4/5	–	–

Edge-function NRMSE (mean over datasets; lower is better); a missed true edge counts as 1. Ratio = BASIL-W / SSC of the cell means. Difference = BASIL-W minus refit with 95% bootstrap interval over datasets. Holm: two-sided exact Wilcoxon signed-rank p, Holm-adjusted over C1–C7 (C8 has 5 datasets, minimum attainable p = 0.0625, not testable), for BASIL-W vs refit and BASIL-W vs SSC. BASIL-W is below SSC in all 75 datasets. Cells share graphs across seeds (C1–C6 use the same 10 DAGs), so they are not independent replications.

3005–3012 (uniform; 8 datasets) and 3005–3009 (Laplace, heteroscedastic; 5 each). Independent-DAG replication: 20 datasets per cell with disjoint seed sets per noise family (Section 6.5). In the replication, graphs and mechanism assignments are shared across cells with the same seed (C1–C6 use the same 10 DAGs; C7 the same DAG structures), so the cells are not independent replications.

Statistics. The dataset is the unit; comparisons are paired by dataset. We report means, 95% percentile bootstrap intervals over datasets (10 000 resamples), exact two-sided Wilcoxon signed-rank tests (the smallest attainable p is 0.002 with 10 pairs and 0.0625 with 5) and Holm adjustment over the seven $d = 100$ cells. Only the registered H1–H5 rules are confirmatory; the rest is exploratory. Pooled tests over cells overstate evidence because graphs repeat.

Compute. All experiments ran on CPU (a shared 4-core machine, one thread per job, at most two jobs in parallel) with PyTorch and NumPy, no GPU. Timings are wall-clock on a loaded machine and are accurate to roughly ±30%.

6 Results

6.1 Structure is found, functions are not: confirmatory replication

Table 2 gives edge-function NRMSE for SSC, refit and BASIL-W on 75 datasets with fresh seeds. SSC's NRMSE is 0.61–0.80 in every cell, while its AUROC is 0.991–0.999 (Table 3): the learner finds the edges but returns curves with a median of 14–46% of the true amplitude (slope; 0.44 in C1). The refit reduces the error to 0.20–0.47 and BASIL-W to 0.15–0.29. BASIL-W has a lower NRMSE than SSC in 75 of 75 datasets, with per-dataset ratios between 0.15 and 0.42, and the cell ratio is 0.22–0.38 (all upper confidence limits ≤ 0.40). Figure 1 shows the dissociation between structure and function; Figure 2 shows the errors by cell; Figure 3 shows recovered curves on one dataset (six true edges drawn at random).

BASIL-W versus the refit. BASIL-W has a lower mean NRMSE than the refit in all seven $d = 100$ cells and the pre-registered tie criterion (BASIL-W $\leq$ refit $+0.03$) holds in 7 of 7. It is significantly better after Holm correction (p = 0.014) only in C2 (Laplace, -0.22), C3 (heteroscedastic, -0.065), C4 (Gaussian, -0.13) and C7 (mechanism family 2, -0.115). In C1, C5 and C6 (uniform noise at $n = 1000$, 500, 2000) the differences are -0.043, -0.030, -0.047, 6–8 of 10 datasets favour BASIL-W, and the supported statement is a tie within about 0.05, not

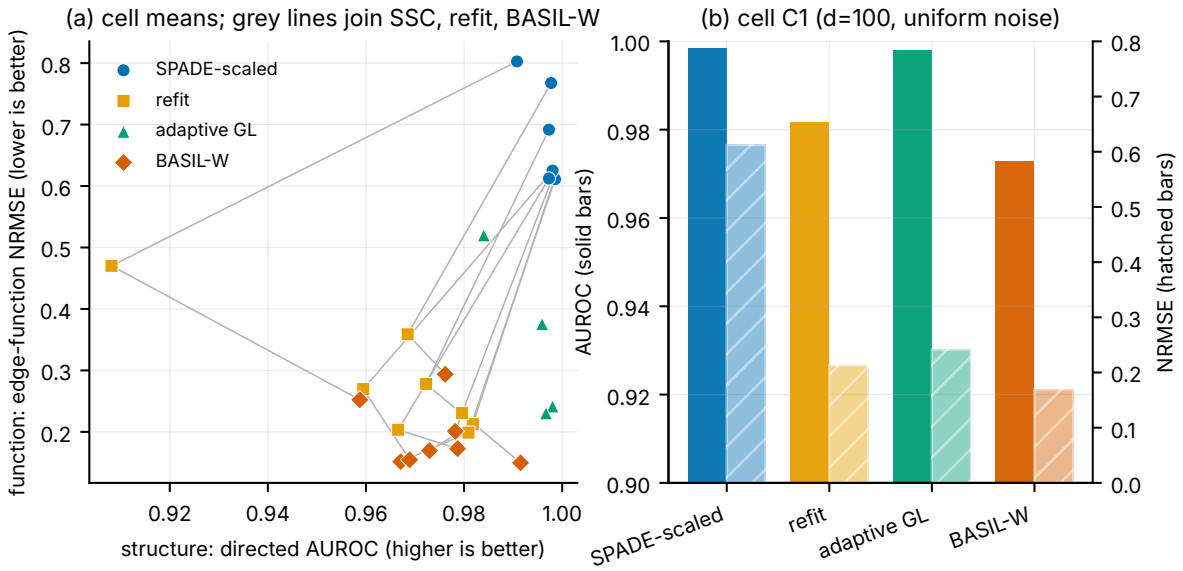


Figure 1: Structure versus function. (a) Cell means of directed AUROC (x) and edge-function NRMSE (y) for SPADE-scaled, refit, adaptive group lasso (only the four cells of the adaptive arm) and BASIL-W; grey lines join SSC, refit and BASIL-W within a cell. The learner with the best structure has the worst functions. (b) Both metrics in cell C1.

Table 3: Structure, interventional-effect error and calibration by cell (means over datasets).

Cell	AUROC SSC	refit	BASIL-W	orient. SSC	refit	BASIL-W	do direct	do indirect	cover.	ECE
C1	0.998	0.982	0.973	0.95	0.92	0.94	0.60 / 0.19 / 0.15	0.86 / 0.26 / 0.22	0.793	0.0014
C2	0.991	0.908	0.959	0.74	0.75	0.92	0.76 / 0.41 / 0.18	0.92 / 0.50 / 0.33	0.753	0.0024
C3	0.998	0.968	0.976	0.89	0.84	0.95	0.73 / 0.32 / 0.20	0.93 / 0.54 / 0.48	0.781	0.0016
C4	0.997	0.972	0.992	0.90	0.88	0.98	0.66 / 0.22 / 0.13	0.87 / 0.34 / 0.23	0.808	0.0008
C5	0.998	0.980	0.978	0.95	0.92	0.95	0.60 / 0.21 / 0.18	0.86 / 0.30 / 0.27	0.798	0.0012
C6	0.999	0.981	0.967	0.95	0.92	0.93	0.60 / 0.16 / 0.15	0.86 / 0.23 / 0.21	0.774	0.0016
C7	0.998	0.959	0.969	0.91	0.90	0.94	0.59 / 0.22 / 0.15	0.82 / 0.31 / 0.27	0.742	0.0015
C8	0.997	0.967	0.979	0.95	0.93	0.95	0.58 / 0.16 / 0.14	0.86 / 0.27 / 0.21	0.788	0.0006

Means over datasets. do-effect NRMSE for up to 20 direct and 20 indirect true pairs per dataset (SSC / refit / BASIL-W). Coverage: pointwise coverage of the 90% BASIL-W band over true edges, kappa = 1; ECE of the edge probabilities. BASIL-W AUROC is lower than SSC in 9 or 10 of 10 datasets in every $d = 100$ cell.

superiority. The refit is fragile under heavy-tailed noise: under Laplace noise its AUROC falls to 0.908 (SSC 0.991) and its NRMSE is 0.47.

Pre-registered hypotheses. Appendix A lists the verdicts. In short: the edge-function (H1) and do-effect (H3) hypotheses were confirmed in all cells; H2 (BASIL-W within 0.03 of the refit) was confirmed; H4 (AUROC loss at most 0.03) missed its target in two cells by 0.002 and 0.001, and the coverage part of H5 failed in seven of eight cells.

6.2 Structure cost and calibration

SSC has the best structure in every cell (AUROC 0.991–0.999). The refit loses 0.016–0.083 AUROC, with the largest loss under Laplace noise. BASIL-W loses 0.006–0.032 relative to SSC (cell means $-0.025, -0.032, -0.022, -0.006, -0.020, -0.031, -0.029, -0.019$ for C1–C8); it is lower than SSC in 9 or 10 of 10 datasets in every $d = 100$ cell (Holm p = 0.014), so the loss is real, and two cells just exceed the 0.03 target. BASIL-W has the best orientation under non-uniform noise (e.g. Laplace 0.919 against 0.745 for SSC). Edge probabilities are well calibrated (ECE ≤ 0.0024). The 90% band coverage is 0.74–0.81 (Table 3; Section 6.8). The cost of BASIL-W is

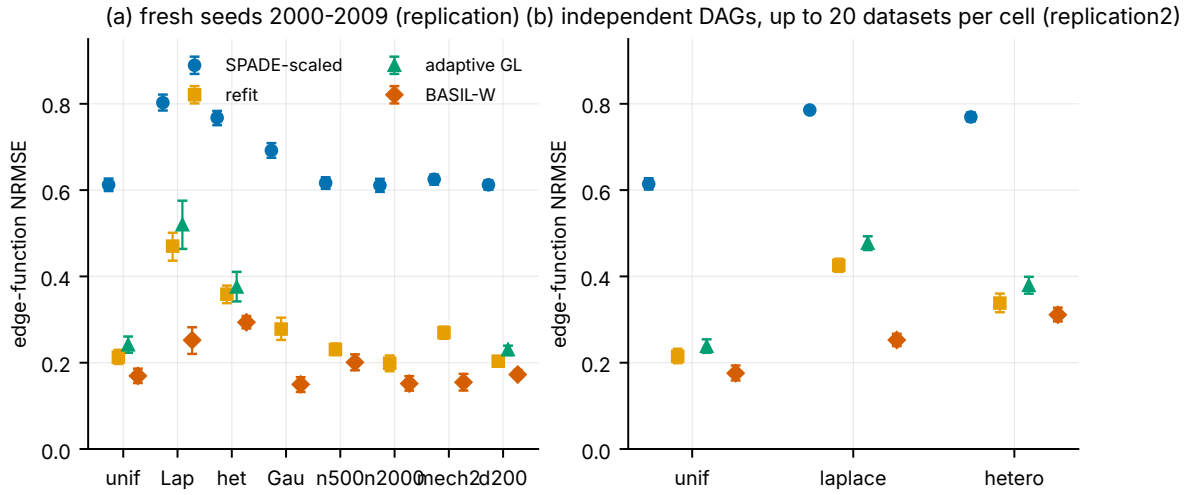


Figure 2: Edge-function NRMSE by cell, with 95% bootstrap intervals over datasets. The adaptive group lasso was run only in C1, C2, C3 and C8 and on a subset of datasets in C2, C3 and C8 (Table 4). A second panel appears when the independent-DAG replication (Section 6.5) has been run.

Table 4: Adaptive group lasso arm (SSA) on the replication seeds, with SSC, refit and BASIL-W taken from the replication runs on the same seeds.

Cell	N	SSC	SSA	refit	BASIL-W	SSA − refit	SSA − BASIL-W	p	AUROC SSA	AUROC BASIL-W
d100 uniform	10	0.612	0.241	0.213	0.169	+0.029 (8/10)	+0.072 (10/10)	0.002	0.998	0.973
d100 Laplace	5	0.805	0.520	0.470	0.261	+0.049 (5/5)	+0.258 (5/5)	0.062	0.984	0.953
d100 hetero	5	0.766	0.375	0.349	0.294	+0.027 (4/5)	+0.082 (5/5)	0.062	0.996	0.977
d200 uniform	3	0.606	0.230	0.193	0.176	+0.037 (3/3)	+0.055 (3/3)	0.250	0.997	0.977

Adaptive group lasso (SSA: weights $(\|\theta\| + 0.01)^{-1}$, same $\lambda = 0.2/d$, γ and ϵ not tuned) on the replication seeds; SSC, refit and BASIL-W values are the replication runs on the same seeds. Differences SSA minus comparator, in parentheses the number of datasets in which SSA is worse. Wilcoxon p (uncorrected) for SSA vs BASIL-W; with 5 or 3 datasets the minimum attainable p is 0.0625 or 0.25.

1.23–1.44 times that of SSC (median wall-clock; SSC 27–31 s and BASIL-W 36–43 s at $d = 100$, $n = 1000$), that of the refit 1.47–1.70 times.

6.3 Interventional effects

Because SSC's curves are shrunken, the interventional-effect curves derived from them are wrong even for direct parents (NRMSE 0.58–0.76) and worse for indirect pairs (0.82–0.93), i.e. errors do not average out along paths but remain large. BASIL-W lowers the direct error by a factor of 3.3–5.3 relative to SSC (ratios 0.19–0.30) and the indirect error by 1.9–4.2 (ratios 0.24–0.52); heteroscedastic noise is the weakest case (indirect ratio 0.52). No dataset has an indirect BASIL-W error above 2 (0 of 75; the maximum is 0.69). Relative to the refit, BASIL-W's do-effects are lower in every cell mean but significantly so only in C2, C3, C4 and C7 for direct pairs (Holm p between 0.014 and 0.039), and only in C4 for indirect pairs (Holm 0.041); in C1, C5, C6 and C8 the two are statistically indistinguishable.

6.4 The standard fix: adaptive group lasso

Before the adaptive arm was run we predicted that SSA would land between SSC and the refit (NRMSE 0.3–0.5). It did better: at $d = 100$ with uniform noise SSA has NRMSE 0.241 (slope 0.90), the same AUROC as SSC (0.998; orientation 0.950), and it is worse than BASIL-W by 0.072 (10 of 10 datasets, p = 0.002) and worse than the refit by 0.029 (8 of 10, p = 0.037). The

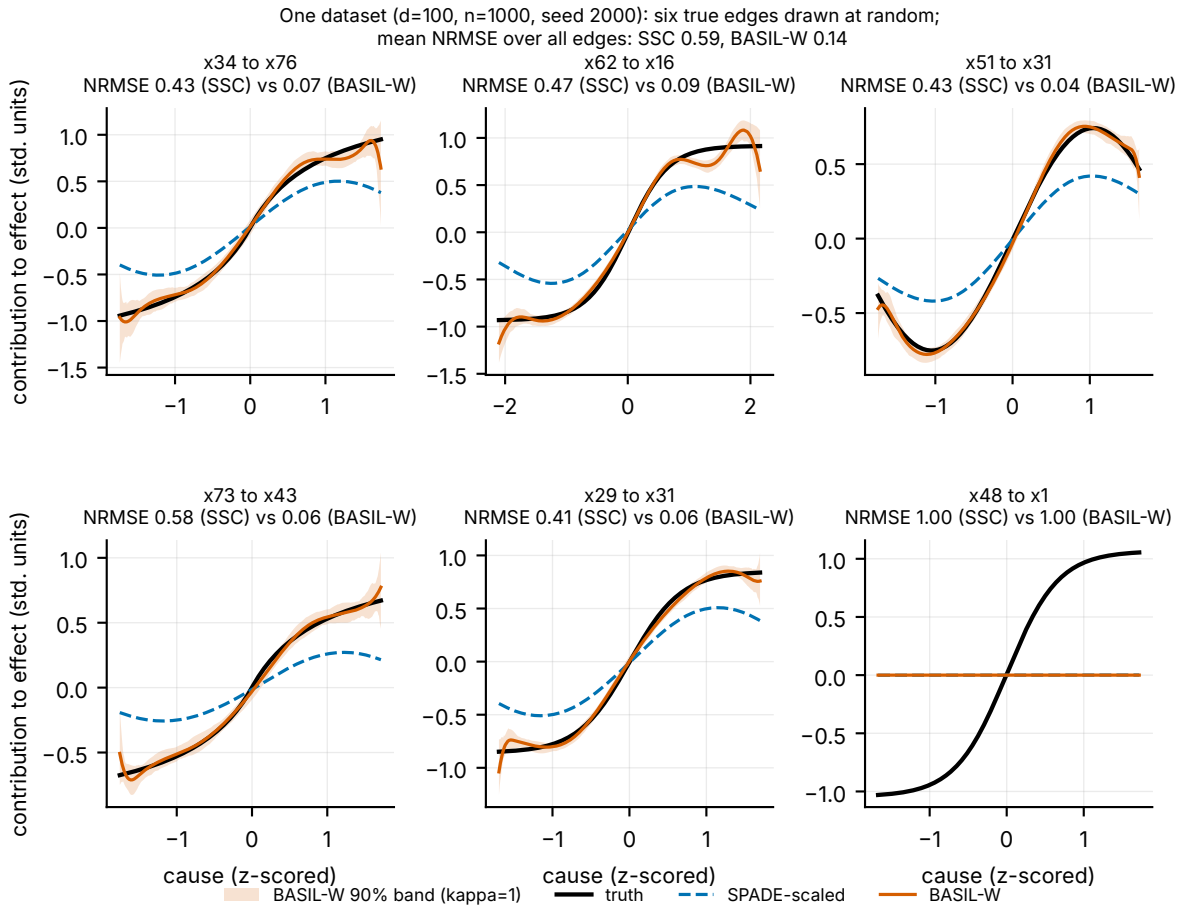


Figure 3: Recovered curves on one dataset ($d = 100$, $n = 1000$, uniform noise, seed 2000). Black: truth; dashed blue: SPADE-scaled; orange: BASIL-W with its 90% band ($\kappa = 1$). The six edges were drawn at random, not selected on any result; the figure illustrates the amplitude shrinkage and, in the last panel, an edge missed by both methods (NRMSE 1); it is not evidence beyond the tables. Data: `data/example_curves.npz`.

SSA values at $d = 200$ (0.230 against BASIL-W 0.176; three datasets), under heteroscedastic noise (0.375 against 0.294) and Laplace noise (0.520 against 0.261) show the same ordering (Table 4). Read plainly: *the standard adaptive group lasso closes about 85% of the NRMSE gap between SPADE-scaled and BASIL-W at no structural cost*, and it is the only repair that preserves SSC's AUROC (BASIL-W loses 0.01–0.03, the refit more). BASIL-W's remaining advantage is small under uniform noise (0.055–0.072) and large under heavy tails (0.26), where the adaptive lasso is also much worse than at light tails. A practitioner who wants accurate curves and the best structure should therefore use the adaptive group lasso unless heavy-tailed noise or edge probabilities matter. The 85% figure is specific to $d = 100$ (about 75% at $d = 500$ on two datasets; Section 6.6). The SSA runs were smaller than the replication (10, 5, 5 and 3 datasets); γ and ϵ were not tuned, and a tuned adaptive lasso could narrow the gap further.

6.5 Independent-DAG replication

The first replication uses 10 datasets per cell and shares graphs between cells. A second pre-registered replication with 20 datasets per cell, independent graphs per noise family and all four arms (SSC, SSA, refit, BASIL-W) is run in `research/replication2`. Table 5 is generated from its results.

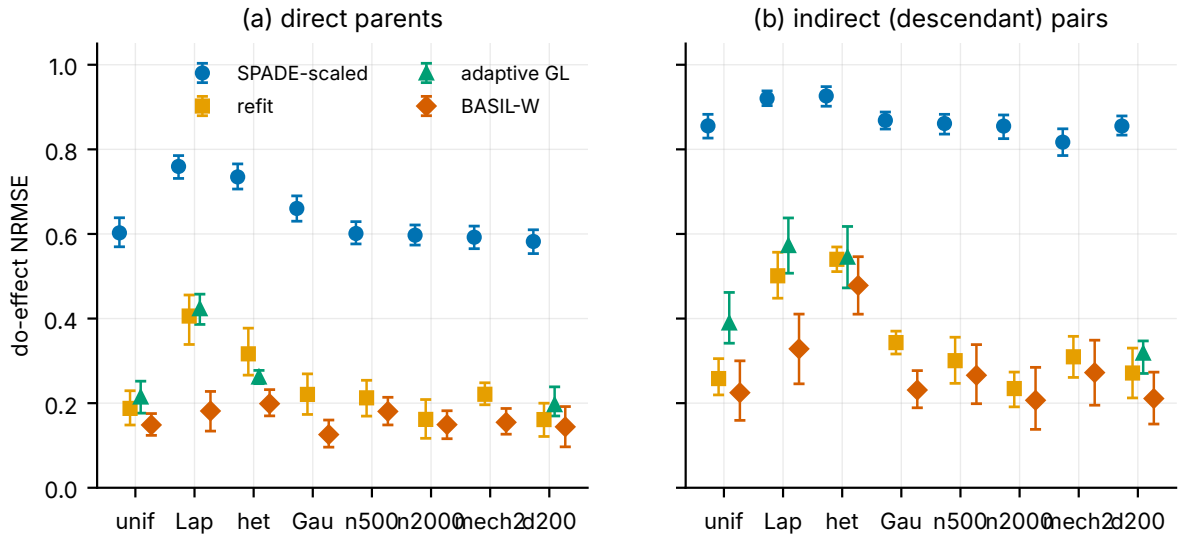


Figure 4: Do-effect NRMSE for direct parents (a) and indirect (descendant) pairs (b), mean and 95% bootstrap interval over datasets, by cell. SPADE-scaled's effects are wrong even for direct parents; the repaired models are much better, and refit and BASIL-W are close.

Table 5: Independent-DAG replication (research/replication2).

Noise	N	SSC	SSA	refit	BASIL-W	BASIL-W − SSA [95% CI]	wins	Holm p	BASIL-W − refit	BASIL-W<SSC
uniform	20	0.614	0.238	0.215	0.176	-0.062 [-0.079, -0.046]	19/20	<0.001	-0.039 (16/20)	20/20
Laplace	20	0.786	0.477	0.426	0.253	-0.224 [-0.239, -0.210]	20/20	<0.001	-0.173 (20/20)	20/20
heteroscedastic	20	0.770	0.379	0.338	0.311	-0.068 [-0.082, -0.054]	19/20	<0.001	-0.027 (14/20)	20/20

Independent-DAG replication (`research/replication2`): complete datasets only (all four arms present), edge-function NRMSE, mean over datasets. BASIL-W − SSA: mean paired difference with 95% bootstrap interval, number of datasets in which BASIL-W is lower, Holm-adjusted exact two-sided Wilcoxon p over the 9 pre-registered NRMSE tests (research/replication2/results/pairwise.csv). BASIL-W − refit: mean difference (wins). Last column: datasets where BASIL-W < SSC. The uniform seeds 3000–3012 coincide with the ablation datasets.

Design. $d = 100$, $n = 1000$, mechanism family 1; 20 datasets per cell and 60 in all, with their own DAGs, mechanism assignments and noise: seeds 3000–3019 (uniform), 3100–3119 (Laplace), 3200–3219 (heteroscedastic). All 60×4 runs completed, with no failures and no dataset with an indirect do-effect NRMSE above 2 for any arm. No parameter was retuned. Statistics as registered: exact Wilcoxon per cell, Holm adjustment over 3 cells × 3 pairs (BASIL-W vs SSC, BASIL-W vs SSA, SSA vs SSC; a separate Holm family for pairs with the refit), bootstrap intervals over datasets.

Verdicts against the plan. **H1 (BASIL-W below SSC in at least 95% of datasets per cell) is confirmed**: 20/20 in all three cells, with NRMSE ratios BASIL-W/SSC of 0.29 (uniform), 0.32 (Laplace) and 0.40 (heteroscedastic). **H2 (BASIL-W lower than SSA) is confirmed** and the kill criterion is not triggered: the SSA minus BASIL-W gaps are 0.062, 0.224 and 0.068, inside the predicted ranges, Holm $p = 1.7 \times 10^{-5}$ in every cell, BASIL-W lower in 19/20, 20/20 and 19/20 datasets. Under uniform noise the gap is small in absolute terms (about 26% of SSA's error) but consistent in sign. **H3 (SSA keeps SSC's AUROC within 0.005; BASIL-W loses 0.01–0.03) is only partly confirmed.** SSA minus SSC AUROC is +0.0001 (uniform) and −0.0005 (heteroscedastic), but −0.0067 under Laplace noise, which misses the 0.005 bound (Holm-significant, SSA lower in 20/20 datasets; small). BASIL-W minus SSC is −0.023, −0.030 and −0.024, inside the predicted range, with the Laplace value at its upper edge;

Table 6: Independent-DAG replication: structure, do-effects, calibration and cost.

Noise	AUROC	do direct	do indirect	cover. (BASIL-W)	ECE	cost ratio
uniform	0.998 / 0.998 / 0.975 / 0.975	0.591 / 0.200 / 0.169 / 0.164	0.849 / 0.378 / 0.278 / 0.262	0.789	0.001	1.58 / 1.57 / 1.34
Laplace	0.994 / 0.987 / 0.926 / 0.964	0.750 / 0.413 / 0.362 / 0.205	0.927 / 0.639 / 0.520 / 0.307	0.761	0.002	1.65 / 1.67 / 1.35
heteroscedastic	0.996 / 0.995 / 0.963 / 0.971	0.723 / 0.281 / 0.260 / 0.214	0.933 / 0.535 / 0.512 / 0.472	0.757	0.002	1.65 / 1.66 / 1.34

Means over complete datasets (SSC / SSA / refit / BASIL-W). Cost: median over datasets of the per-dataset cost ratio to SSC (SSA / refit / BASIL-W), excluding metric code, shared loaded machine.

BASIL-W is lower than SSC in 20/20, 20/20 and 19/20 datasets. **H4 (do-effect NRMSE BASIL-W $\leq$ SSA $<$ SSC) is met on the cell means** in all cells for direct and indirect pairs, but the BASIL-W advantage over SSA is not established everywhere: for direct effects under uniform noise the difference is -0.037 [-0.074, $+0.007$], Holm $p = 0.07$ (14/20 datasets favour BASIL-W), and for indirect effects under heteroscedastic noise it is -0.063 [-0.119, $+0.006$] with the interval including 0 (Holm $p = 0.014$). Both are ties in practice. SSA is below SSC in every dataset and cell (20/20). H5 (cost, descriptive): median cost relative to SSC is 1.65, 1.65, 1.69 for SSA, 1.62, 1.69, 1.66 for the refit and 1.34, 1.35, 1.35 for BASIL-W (uniform, Laplace, heteroscedastic); on a loaded shared machine these ratios are $\pm$30% estimates.

BASIL-W against the refit is weak outside heavy tails. The pooled picture over the 60 datasets (exploratory; graphs are independent across cells here) is BASIL-W minus SSA -0.118 [-0.139, -0.098], BASIL-W lower in 58/60; BASIL-W minus refit -0.080 [-0.099, -0.061], lower in 50/60. By cell, BASIL-W beats the refit by only 0.039 under uniform noise and 0.027 under heteroscedastic noise (Holm $p = 0.003$ and 0.008), and the refit has the lowest NRMSE of the four arms on 4/20 uniform and 6/20 heteroscedastic datasets (BASIL-W 15/20 and 14/20); only under Laplace noise is BASIL-W clearly better (0.173, 20/20). The adaptive group lasso is consistently a little worse than the plain refit on edge NRMSE (SSA minus refit $+0.023$, $+0.051$, $+0.041$; all Holm-significant). In cell means the adaptive group lasso closes about 85% of the SSC-to-BASIL-W NRMSE gap in the uniform and heteroscedastic cells and 58% under Laplace noise. These results do not support calling BASIL-W the best repair: it is the best of the four arms on average, with the advantage over the refit being practically small under light-tailed or heteroscedastic noise.

Comparison with the first replication. Everything reproduces qualitatively on independent DAGs: signs, orderings and the structure result that SSA keeps SSC's AUROC. The BASIL-W advantage is somewhat smaller on the new data. The SSA gap shrank by 0.01–0.03 (0.062/0.224/0.068 against 0.072/0.258/0.082 earlier), and the BASIL-W minus refit difference under heteroscedastic noise fell from -0.065 (5 earlier datasets) to -0.027. Under uniform noise it is nearly unchanged (-0.039 against -0.043) and now significant after Holm correction. The earlier concern that the first replication reused DAGs across cells is addressed for the three noise families here, and the headline conclusions did not depend on those 10 graphs.

Calibration. BASIL-W's 90% bands cover 0.789, 0.761 and 0.757 of the true curve (uniform, Laplace, heteroscedastic) against the nominal 0.90, so the under-coverage found earlier persists; ECE of the edge probabilities is 0.0014–0.0021.

Caveats. This is a replication of the experiment with the same generator, code family and hyperparameters, not an external validation; the three cells share mechanism family and density. The uniform seeds 3000–3012 coincide with the ablation validation and test datasets of Sections 6.7 and 6.8; no BASIL-W setting was changed by that work, but the uniform cell is not fully fresh. The adaptive group lasso used the untuned $\gamma = 1$, $\epsilon = 0.01$, which is the largest open threat to the BASIL-W-over-SSA result (Section 6.6). Direct do-effects of BASIL-W do not differ

Table 7: Adaptive arm at $d = 500$ and (γ, ϵ) sensitivity (research/adaptive2).

(a) $d = 500$ (individual datasets and mean).

Seed	Method	NRMSE	slope	AUROC	orient.	do direct	time (s)
1000	SSC	0.693	0.31	0.999	0.920	0.73	472
1000	refit	0.229	0.93	0.955	0.912	0.28	696
1000	SSA	0.308	0.84	0.999	0.918	0.27	726
1000	BASIL-W	0.166	1.00	0.974	0.947	0.13	540
1001	SSC	0.686	0.34	0.997	0.917	0.63	471
1001	refit	0.228	0.94	0.951	0.911	0.10	730
1001	SSA	0.282	0.88	0.996	0.919	0.12	717
mean	SSC	0.690	0.33	0.998	0.919	0.68	–
mean	refit	0.229	0.93	0.953	0.912	0.19	–
mean	SSA	0.295	0.86	0.997	0.918	0.19	–
mean	BASIL-W	0.166	1.00	0.974	0.947	0.13	–

Two datasets: no significance claims.

(b) (γ, ϵ) grid, $d = 100$, uniform, seeds 2000–2004.

γ	ϵ	N	NRMSE	slope	AUROC
0.5	0.001	5	0.228	0.89	0.998
0.5	0.01	5	0.318	0.79	0.998
0.5	0.1	5	0.359	0.77	0.998
1	0.001	5	0.199	0.92	0.999
1	0.01	5	0.234	0.89	0.998
1	0.1	5	0.336	0.78	0.998
2	0.001	5	0.215	0.90	0.999
2	0.01	5	0.204	0.92	0.999
2	0.1	5	0.245	0.90	0.998

Reference on the same seeds (replication): SSC 0.610, refit 0.216, BASIL-W 0.159. The default $(1, 0.01)$ is the configuration used for all headline numbers; the grid is a sensitivity analysis and no setting was selected.

detectably from SSA under uniform noise (Holm $p = 0.07$) or from the refit under uniform noise (Holm $p = 0.43$). The datasets in a cell are independent, but there are only three noise families. Nothing is concluded here about real data, other generators, mechanism family 2, other n or d, Gaussian noise or $d \geq 200$. The plan had no deviations; one crash in `aggregate.py` (fewer than 3 complete datasets in a cell) was fixed without changing any statistic.

6.6 Adaptive arm at $d = 500$ and sensitivity to γ and ϵ

The adaptive arm was run only up to $d = 200$ in the main grid. A pre-registered extension (`research/adaptive2`) runs it at $d = 500$ (two datasets) and over a 3×3 grid of (γ, ϵ) at $d = 100$ (five datasets) to test whether the gap to BASIL-W is an artefact of the untuned defaults.

Part A: $d = 500$ (two datasets, descriptive). With $n = 1000$, uniform noise, mechanism family 1 and seeds 1000 and 1001, the mean edge-function NRMSE is 0.690 for SSC, 0.295 for SSA ($\gamma = 1$, $\epsilon = 0.01$), 0.229 for the refit and 0.162 for BASIL-W (per seed SSA 0.308 and 0.282, BASIL-W 0.166 and 0.157). AUROC is 0.9983 (SSC), 0.9975 (SSA), 0.9534 (refit) and 0.9773 (BASIL-W); orientation 0.919, 0.919, 0.912 and 0.952. Re-run SSC and refit reproduce the earlier CSVs exactly, so the comparison is on identical models. The pre-registered predictions fared as follows. P1 (SSA NRMSE 0.22–0.30, next to the refit, worse than BASIL-W): inside the

numerical range but at its top; SSA is 0.066 worse than the refit and 0.133 worse than BASIL-W, a larger gap than at $d = 100$ and 200 (+0.072 and +0.055 against BASIL-W), so SSA is not next to the refit. P2 (AUROC within 0.005 of SSC): met. **P3 (SSA do-effects between the refit and BASIL-W): not met.** Direct effects tie with the refit (0.19 against 0.19) and indirect effects are clearly worse (0.42 against 0.28 for the refit and 0.17 for BASIL-W), though far better than SSC (0.90). SSA's do-effects are therefore worse than the plain refit's, a failed prediction. P4 (fit time near the refit): met (about 250 s against 225–259 s). At $d = 500$ the adaptive group lasso closes about 75% of the SSC-to-BASIL-W gap, $(0.690 - 0.295)/(0.690 - 0.162)$, against about 85% at $d = 100$; the sentence about 85% is specific to $d = 100$. BASIL-W's fit time beyond the shared SPADE-scaled fit is smaller (69 s, one dataset) than either refit; BASIL-W seed 1001 values come from an earlier independent run on the same data.

Part B: (γ, ϵ) grid (five datasets, sensitivity only). At $d = 100$, uniform noise, seeds 2000–2004, the mean NRMSE over the nine settings ranges from 0.199 to 0.359, a spread of 0.16 (the default 0.234, per-dataset 0.210, 0.191, 0.211, 0.297, 0.260). The registered expectation of a spread below 0.05 (Q1) failed: $\epsilon = 0.1$ is clearly worse (0.245–0.359) at every γ, consistent with a large ϵ flattening the weights towards a plain group lasso; the cell $(0.5, 0.01)$ at 0.318 is worse than both its neighbours in ϵ and the pattern is not smooth in γ, which five datasets cannot resolve. AUROC is 0.9978–0.9988 in all nine settings, so structure is insensitive to γ and ϵ; the expectation that $\gamma = 2$ with small ϵ loses AUROC failed (it has the highest, 0.9988). No setting reaches BASIL-W (0.159 on these seeds; rule: at most 0.169): the best grid mean, $(1, 0.001)$ at 0.199, is still 0.040 above (Q3 confirmed). Two settings beat the default by more than 0.03 on the mean and on 5/5 datasets, $(1, 0.001)$ by 0.035 and $(2, 0.01)$ by 0.030. By the pre-registered rule they are "clearly better" in this sample, but they were examined on the same seeds used for the sensitivity analysis, and the default had already been run on seeds 2000–2004. **They are not adopted**: the default $(1, 0.01)$ remains the SSA configuration for every headline number. If an improved setting is wanted, the two candidates should be validated against the default on fresh seeds with the choice and threshold written first.

What this changes. The statement that γ and ϵ were untuned can now be accompanied by a sensitivity result: structure does not depend on them, edge-function error does (0.20–0.36, mainly through ϵ), and a smaller ϵ or $\gamma = 2$ would likely shrink SSA's gap to BASIL-W at $d = 100$ by roughly a quarter to a third without closing it. The claim that BASIL-W has lower NRMSE than the adaptive group lasso holds in every grid setting at $d = 100$ (on means), but this does not remove the tuning threat, because the grid is five datasets, one noise family and one mechanism family, and the grid seeds cannot select a setting.

Limitations. Two datasets at $d = 500$ (BASIL-W time from one), five for the grid, no significance claims, one generator family, uniform noise and mechanism family 1 only; wall-clock times on a loaded shared machine.

6.7 Ablations

Table 8 summarises the ablations. The warm start from SSC's order is essential: a cold start raises NRMSE from 0.177 to 0.324 and lowers orientation from 0.943 to 0.809 (8 of 8 datasets worse), because cold chains do not converge in $200d$ iterations (end-score gaps of hundreds of nats); longer cold chains were not run, so this is a statement about this chain length. The source of the candidate set does not matter at $d = 100$ (candidate recall is 1.000 with either ranking), and the prior scale g and inclusion prior π are robust ($|\Delta\text{NRMSE}| \leq 0.002$). Capping the parent set at $k_{\max} = 2$ is harmful (+0.103 NRMSE, 8 of 8 worse; with mean degree 2 a non-trivial share of nodes have three or more parents). Four chains of $600d$ iterations change NRMSE by

Table 8: BASIL-W ablations: one change at a time ($d = 100$, $n = 1000$, uniform noise, 8 test datasets).

Variant	NRMSE	Δ	orient.	Δ	AUROC	do indirect	cover.	gap	time
BASIL-W (reference)	0.177	–	0.943	–	0.971	0.248	0.798	8.8	10.8
candidates by marginal R^2	0.176	-0.001	0.943	+0.000	0.972	0.251	0.798	17.8	10.6
cold start (random order)	0.324	+0.147	0.809	-0.133	0.909	0.548	0.678	543.0	9.6
cold start + R^2 candidates (BASIL-S)	0.318	+0.141	0.821	-0.121	0.921	0.593	0.697	575.9	10.0
$g = n/4$	0.176	-0.001	0.945	+0.002	0.972	0.251	0.801	15.7	10.3
$g = 4n$	0.176	-0.001	0.942	-0.000	0.974	0.253	0.799	65.6	10.7
$\pi \times 0.5$	0.177	-0.000	0.943	+0.000	0.972	0.248	0.798	7.5	10.0
$\pi \times 2$	0.176	-0.002	0.945	+0.002	0.973	0.251	0.800	17.1	10.3
$k_{max} = 2$	0.280	+0.103	0.894	-0.048	0.951	0.422	0.706	85.4	6.9
4 chains, $600d$ iterations	0.173	-0.004	0.934	-0.009	0.974	0.227	0.807	30.9	23.5

d=100, n=1000, uniform noise, 8 test datasets (seeds 3005–3012), one change at a time. Δ = paired mean difference to the reference. End-score gap: max−min over chains of the mean log-score at the end of the chains (nats); time: BASIL fit only, 1 core, seconds. No significance tests.

Table 9: Band coverage before and after variance inflation, test datasets.

Noise (datasets)	κ	all edges	found	missed	whole curve	mean width
uniform (8)	1	0.798	0.877	0.095	0.15	0.140
uniform (8)	1.55	0.894	0.981	0.112	0.61	0.216
Laplace (5)	1	0.796	0.850	0.207	0.05	0.141
Laplace (5)	1.55	0.904	0.964	0.249	0.32	0.219
hetero (5)	1	0.790	0.835	0.304	0.03	0.174
hetero (5)	1.55	0.908	0.960	0.357	0.23	0.269

Pointwise coverage of the 90% band (5th/95th percentile of $m + \kappa(D - m)$ for posterior draws D and mean m) over true edges, mean over datasets (found/missed: pooled over edges). $\kappa^* = 1.55$ was chosen as the smallest value reaching coverage 0.90 on 5 validation datasets (seeds 3000–3004) and frozen. 'found' = posterior inclusion $P > 0.9$; 'missed' = $P \leq 0.9$; 'whole curve' = share of curves entirely inside the band.

−0.004 and indirect do-effect NRMSE from 0.248 to 0.227, but the chain end-score gap does not shrink (31 nats against 8.8), so the chains are not provably converged; the default is adequate for edge-function NRMSE and not demonstrably so for indirect effects. Variants that keep the warm start all look alike, so BASIL-W's output here is largely determined by SSC's order and candidates plus local refinement; what happens when SSC's order is poor was not tested. Three registered predictions were wrong (the size of the warm-start effect, the effect of $k_{\max} = 2$ and chain convergence).

6.8 Credible bands

On five validation datasets the bands at $\kappa = 1$ cover 0.802 of the truth pointwise (nominal 0.90); only 11% of the curves lie entirely inside the band. The diagnosis is that 72% of the shortfall comes from the 8% of true edges the model does not find ($P \leq 0.9$): their band is a spike of width 0.03 around zero while the truth is not zero. For found edges the shortfall is small (0.87 against 0.90), z-scores are centred (mean −0.004, sd 1.03), and the noise variance, shrinkage and spline-approximation error are ruled out as main causes. We therefore tried a single remedy, posterior-variance inflation $m + \kappa(D - m)$. The smallest κ with validation coverage ≥ 0.90 is $\kappa^* = 1.55$ (frozen), which gives test coverage 0.894, 0.904 and 0.908 for uniform, Laplace and heteroscedastic noise with bands 1.55 times wider (Table 9). This is average validity obtained partly for the wrong reason: found edges are over-covered (0.96–0.98 instead of 0.90) while missed edges remain uncovered (0.11–0.36), and the whole-curve (simultaneous) coverage is only 0.61 (uniform). The per-dataset coverage ranges from 0.78 to 0.93 for the uniform test datasets at

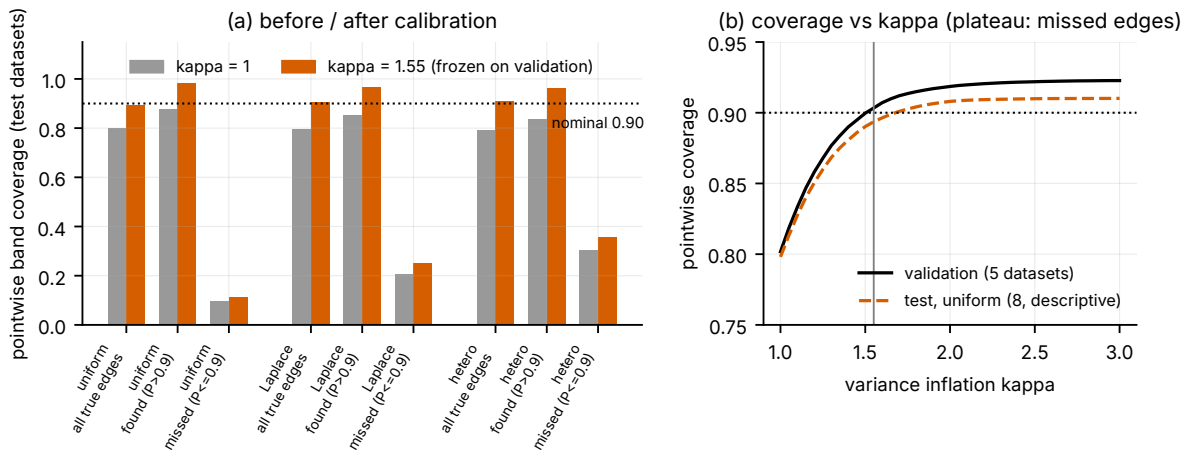


Figure 5: Coverage of 90% bands before ($\kappa = 1$) and after ($\kappa = 1.55$, frozen on validation) calibration. (a) By noise family and by whether the edge was found; (b) coverage as a function of κ on validation and, for description only, on the uniform test datasets.

Table 10: Semi-synthetic Sachs benchmark (research/semisyn).

d	n	Noise	N	SSC	SSA	refit	BASIL-W	GAM-ridge	ratio	AUROC SSC / BASIL-W	cover.
11	2000	bootstrap residuals	20	0.826	0.711	0.730	0.391	0.394	0.47	0.821 / 0.893	0.52
11	2000	Gaussian	20	0.714	0.495	0.516	0.283	0.362	0.40	0.860 / 0.993	0.63
11	500	bootstrap residuals	20	0.835	0.741	0.759	0.652	0.539	0.78	0.817 / 0.837	0.39
11	500	Gaussian	20	0.721	0.563	0.586	0.585	0.530	0.81	0.856 / 0.965	0.45
110	2000	bootstrap residuals	5	0.930	0.820	0.757	0.398	–	0.43	0.957 / 0.919	0.53

Semi-synthetic Sachs benchmark (research/semisyn; 17 true edges on 11 nodes, ten copies at $d = 110$): edge-function NRMSE, mean over datasets. Ratio: mean of per-dataset BASIL-W/SSC. Coverage: 95% band of BASIL-W over true curves. GAM-ridge did not finish at $d = 110$. At $d = 11$ SSC and plain SPADE are identical. The truth is a fitted model of contested data.

κ^* because the share of missed edges was 16% on test and 8% on validation; κ^* was somewhat lucky and about 1.7–2.0 would have been needed on the test sets to reach 0.90. We conclude that edge probabilities (ECE ≤ 0.0024) are calibrated, curve bands are not conditionally valid, and that the honest remedy for missed edges is a better structure estimate. Split-conformal and bootstrap alternatives were not implemented.

6.9 Semi-synthetic benchmark from Sachs

Real data have no function ground truth. A semi-synthetic design fits a model to the Sachs data and simulates from the fitted SEM, so that edge functions are known while the graph and the functional shapes are of empirical origin (`research/semisyn`).

Construction. The true structure is the Sachs consensus graph with one edge removed: the 18-edge graph contains the directed cycle PIP2 $\to$ PIP3 $\to$ plcg $\to$ PIP2, and the weakest cycle edge by endpoint correlation, PIP3 $\to$ plcg ($r = 0.03$), was dropped by a rule fixed beforehand, leaving **17 true edges** on 11 nodes. For each node an additive cubic B-spline model with ridge penalty chosen by cross-validation was fitted to all 7466 log-transformed, z-scored rows; these 17 fitted functions are the ground truth. Noise is either a bootstrap of each node's fitted residuals ("bootstrap", real, heavy-tailed and skewed) or Gaussian with the same standard deviation. The fitted model has median node R^2 0.33, true-curve amplitudes from 0.05 to 0.79 standard deviations (9 of 17 edges with amplitude at least 0.3) and non-linear shares from 0.08 to 0.97. At $d = 110$ there are ten disjoint copies (170 true edges, columns shuffled), bootstrap noise,

$n = 2000$, five datasets. In all, 85 of 85 datasets finished with none dropped (seeds 3500–3819 at $d = 11$, 5000–5004 at $d = 110$). At $d = 11$, SSC and plain SPADE are bit-identical in all 80 datasets, because the screening is vacuous (K=15 exceeds $d - 1$) and $\lambda = \min(0.01, 0.2/d) = 0.01$, so every $d = 11$ statement about SPADE-scaled is a statement about plain SPADE; at $d = 110$ plain SPADE collapses (NRMSE 1.00, AUROC 0.58) while SSC keeps structure (0.957).

Results. BASIL-W lowers the NRMSE of SSC in all 80 $d = 11$ datasets and all 5 $d = 110$ datasets, but the size of the gain depends on n. The BASIL-W/SSC ratio is 0.47 [0.455, 0.490] (bootstrap noise) and 0.40 (Gaussian) at $n = 2000$, 0.78 and 0.81 at $n = 500$, and 0.43 at $d = 110$. At $n = 500$ the gain comes from strong edges (amplitude at least 0.3; ratios 0.57 and 0.44) while weak edges are not improved (0.94 and 1.03): BASIL-W drops them. SSC is heavily shrunk here (median slope 0.01–0.22, far below the iSCM simulations' 0.14–0.46). The standard fixes help less than in the simulations: SSA improves on SSC by only 0.09–0.22 NRMSE because re-weighting cannot recover edges SSC has already shrunk to zero (58–73% of true edges at $n = 2000$), and the refit ($\text{SSR}_{k=1.5}$ keeps only 16 edges where 17 are true, with AUROC down to 0.70) is only 0.08–0.20 below SSC. SSA minus BASIL-W is positive in four of five settings (+0.320, +0.211, +0.089, and +0.423 at $d = 110$; the exception is $n = 500$ Gaussian, −0.022 [−0.044, +0.001]). BASIL-W's band coverage of the true curve is 0.39 ($n = 500$ bootstrap), 0.45, 0.52, 0.63 ($n = 2000$ Gaussian) and 0.53 ($d = 110$), against 0.74–0.81 in the simulations; coverage is below the plan's 0.60 line in three of five settings. Part of the reason is that 23–48% of true edges are dropped under bootstrap noise, so their bands collapse, and part that the true functions are fitted splines with some wiggle.

Agreement with the simulation story is only partial. By the rule stated in the plan (BASIL-W/SSC at most 0.7 with the interval's upper limit below 0.8 in at least three of four $d = 11$ cells and at $d = 110$), the main prediction P3 is **partial**: it holds in two of four $d = 11$ cells and at $d = 110$. The repair headline replicates in direction in all five settings (SSC worst, BASIL-W best among the SPADE-family arms), but in size only at $n = 2000$. The parts specific to the structure cost, the refit comparison and calibration do not replicate cleanly. (i) BASIL-W has higher AUROC than SSC at $d = 11$ (by 0.02 to 0.13, with much higher AUPRC and orientation), the opposite sign to the simulations; at $d = 110$ the simulation pattern returns (−0.037 [−0.051, −0.028], 5/5 datasets lower). (ii) The refit is not close to BASIL-W as it is at $d = 100$: BASIL-W is better by 0.34 and 0.23 at $n = 2000$, 0.107 at $n = 500$ bootstrap, and tied at $n = 500$ Gaussian (−0.002). (iii) Do-effect ratios BASIL-W/SSC are 0.58, 0.42, 0.81, 0.80, 0.45 (direct) and 0.85, 0.26, 1.01, 0.85, 0.70 (indirect); at $n = 500$ bootstrap there is no indirect gain. (iv) SSC's AUROC at $d = 11$ is only 0.82–0.86, so the simulation claim that structure is almost perfect does not hold here. (v) Calibration of the bands collapses (see above).

A plain GAM is a competitor at $d = 11$. GAM-ridge, a ridge-penalised additive regression on all parents without a DAG constraint, ties BASIL-W on edge NRMSE at $n = 2000$ with bootstrap noise (0.394 against 0.391, paired $p = 0.96$) and is better at $n = 500$ (0.539 against 0.652, GAM lower in 20/20 datasets; 0.530 against 0.585 under Gaussian noise). BASIL-W is better at $n = 2000$ with Gaussian noise (0.283 against 0.362). GAM-ridge has worse structure and orientation (AUPRC 0.53–0.59, orientation 0.56–0.77). It did not finish at $d = 110$ (more than 60 min of CPU on one dataset, against 14 min for all other arms), so it is infeasible at the scale the pipeline targets; no GAM-ridge comparison is made on the $d = 100$ iSCM data. Other baselines: NOTEARS-linear reaches AUROC 0.73–0.74 at $d = 11$ and chance-level AUPRC at $d = 110$; var-sortnregress 0.60–0.70.

The real-data pattern is not reproduced. On the real Sachs data SSC has AUROC 0.613 and BASIL-W 0.563 (Section 6.11); here the same graph and fitted functions with iid data give

SSC 0.82–0.86 and BASIL-W 0.84–0.99, BASIL-W above SSC. Whatever limits the real-data scores (pooled interventional conditions, the contested graph, hidden variables, non-additivity) is therefore not captured by estimator behaviour on this generator, and conversely this benchmark says nothing about the selection instability seen on real data, because subsample stability was not measured here. Real-data amplitude ratios (refit/SSC 1.60, BASIL-W/SSC 1.41) are much smaller than the shrinkage visible here, though they are computed on the top-ranked edges only.

Caveats. The truth is a fitted model of contested real data: the consensus graph is contested, contains a cycle (one edge dropped by our rule) and omits known relations; results concern estimators on a generator with real-data-derived functions, not Sachs biology. Many fitted edges are weak or nearly linear (six of 17 below amplitude 0.25), and near-linear edges are weakly oriented under additive-noise identifiability; BASIL-W drops plcg → PIP2 (NRMSE 1.00) where GAM-ridge reaches 0.16, probably an orientation ambiguity (not checked). Ridge-CV picked very small penalties for several nodes, so some true functions are wiggly fits. The four $d = 11$ cells share one SEM, so they are not replications of the construction; $d = 110$ is ten identical copies with five datasets (minimum attainable exact Wilcoxon $p = 0.0625$, only effect sizes and intervals are interpretable). The truth uses the same B-spline family as the spline learners, which favours them. Hyperparameters were not tuned for this generator. The cost column includes metric code and is not comparable with Section 6.1. One post-registration code change (a fixed CSV column set) did not alter any metric; GAM-ridge was dropped at $d = 110$ after exceeding the time limit.

6.10 External baselines at small d

This section compares our estimators with published nonlinear DAG learners, using the registered plan `research/external/PLAN.md` (commit c245cc1, before any run on the evaluation seeds). It was added after the other experiments and changes the reading of the earlier ones. Setting: iSCM, mechanism family 1, $n = 1000$, three cells of 10 datasets each (seeds 4000–4009): $d = 10$ uniform (E1), $d = 10$ Laplace (E3), $d = 20$ uniform (E2); 13 arms × 30 datasets = 390 runs, no errors. **Only $d \leq 20$ was tested**; the large-d regime for which SPADE-scaled and BASIL-W were designed was not, and CAM-like was not run at $d \geq 50$.

Arms. Ours: SSC, SSA, refit (SSR, $k = 1.5$), BASIL-W (the pre-registered best arm by pooled TOPK NRMSE) and GAM-ridge as a reference. External: NOTEARS-MLP (repository version) and a port of the original (NOTEARS-MLP-gc), DAGMA-nonlinear (repository call, 4000/8000 iterations rather than the library defaults of 50000/80000), DCE-like (derivative-based post-hoc score on the DAGMA-MLP fit, so its curves are identical to DAGMA-nonlinear's), GraN-DAG (gcastle defaults, without CAM pruning), **CAM-like** and two NAM-parameterised SPADE variants. CAM-like is **our own Python reimplementation of the CAM recipe** (greedy forward edge addition with a BIC-penalised log-RSS gain, OLS B-spline terms, F-test pruning at $p < 0.001$), not the original R code, which could not be run; the NAM variants are not the original NAM and were untuned. Dagma-DCE proper could not be run. Edge-function NRMSE is scored under the TOPK rule (every method receives the oracle edge count $k = |E|$, missed edges count 1), which is the pre-registered primary; the own-graph and no-selection rules give the same conclusions.

An external method beats ours in part of the grid. CAM-like has a lower edge-function NRMSE than BASIL-W under Laplace noise at $d = 10$: 0.151 against 0.281 (paired difference 0.130, 95% interval [0.061, 0.218], Wilcoxon $p = 0.002$, CAM-like better on 10/10 datasets). At $d = 20$ uniform it is lower, 0.155 against 0.211, but not significantly (interval [−0.005, 0.110], $p = 0.084$); against the best of our arms in that cell, SSA at 0.172, the gap is 0.017, a tie. At

Table 11: External baselines against our estimators, $d \leq 20$ (research/external).

Method	NRMSE E1	E3	E2	pooled	AUROC E1	E3	E2	pooled
SSC	0.372	0.617	0.564	0.518	0.994	0.969	0.988	0.984
SSA	0.167	0.449	0.172	0.263	0.995	0.959	0.992	0.982
refit	0.167	0.405	0.203	0.258	0.990	0.927	0.974	0.964
BASIL-W	0.144	0.281	0.211	0.212	0.989	0.940	0.932	0.954
CAM-like (own Python)	0.176	0.151	0.155	0.161	0.954	0.985	0.959	0.966
GAM-ridge	0.547	0.466	0.591	0.535	0.939	0.955	0.969	0.954
NAM-refit (untuned)	0.344	0.313	0.420	0.359	0.874	0.906	0.837	0.873
NOTEARS-MLP	0.813	0.786	0.879	0.826	0.685	0.680	0.658	0.674
NOTEARS-MLP-gc	0.843	0.782	0.757	0.794	0.621	0.707	0.705	0.678
DAGMA-nonlinear	0.705	0.724	0.737	0.722	0.590	0.594	0.557	0.580
DCE-like	0.692	0.724	0.745	0.720	0.676	0.653	0.679	0.669
GraN-DAG	0.745	0.749	0.937	0.810	0.670	0.690	0.542	0.634

Edge-function NRMSE under the TOPK rule (oracle edge count; missed true edges count 1) and directed AUROC; mean over 10 datasets per cell (seeds 4000–4009), iSCM, mechanism family 1, $n = 1000$. Pooled = 30 datasets that are not independent (E1 and E3 share DAGs). CAM-like is our own Python reimplementation of the CAM recipe, not the original R code. External methods were run with default or repository settings and untuned.

Paired difference BASIL-W − CAM-like (TOPK NRMSE; positive = CAM-like better).

Cell	N	mean diff [95% CI]	Wilcoxon p (uncorrected)	BASIL-W lower
$d = 10$ unif.	10	-0.031 [-0.083, +0.014]	0.625	4/10
$d = 10$ Lap.	10	+0.130 [+0.061, +0.219]	0.002	0/10
$d = 20$ unif.	10	+0.056 [-0.005, +0.109]	0.084	2/10
pooled (30)	30	+0.052 [+0.010, +0.097]	0.013	6/30

$d = 10$ uniform it is slightly worse, 0.176 against 0.144 (difference 0.031, interval $[-0.014, 0.083]$, $p = 0.625$; CAM-like better on 6/10 datasets), which the registered rule labels inconclusive and which is practically a tie. Pooled over the 30 datasets CAM-like is at 0.161 and BASIL-W at 0.212 (difference 0.052, interval [0.010, 0.096], $p = 0.013$), but the 30 datasets are not independent because E1 and E3 share DAGs, so the pooled test overstates the evidence. The registered prediction that CAM-like would match or beat our best arm in at least two of three cells was met. CAM-like also ties on structure (pooled AUROC difference -0.012 $[-0.041, 0.012]$; BASIL-W is better at $d = 10$ uniform by 0.035, CAM-like at Laplace by 0.045 with an interval touching 0) and on do-effects (pooled direct 0.168 against 0.145 and indirect 0.281 against 0.236 for BASIL-W and CAM-like, both intervals including 0), and its amplitude is calibrated (median slope 0.99–1.00, BASIL-W 0.93–1.00). *Our family therefore does not win a head-to-head against a well-specified additive spline learner at $d \leq 20$ on this benchmark.* The expected weakness of CAM-like under Laplace noise did not appear (orientation 0.97 against 0.92 for BASIL-W).

Neural learners are far behind, untuned. The MLP-based learners (NOTEARS-MLP, NOTEARS-MLP-gc, DAGMA-nonlinear, DCE-like, GraN-DAG) have TOPK NRMSE 0.69–0.94 per cell (pooled 0.72–0.83), AUROC 0.54–0.71 against 0.93–0.99 for the spline-based arms, pooled orientation 0.45–0.59 (near chance), and median calibration slopes of 0.00–0.28 (a missed edge counts as slope 0). Differences against our best arm are +0.51 to +0.61 NRMSE pooled, $p = 0.002$ (the smallest attainable with 10 pairs) in every cell. Their published evaluations are structural, and here they ran with default or repository settings and were not tuned on this generator, whereas the SPADE rule $\lambda = 0.2/d$ and $k = 1.5$ were chosen on earlier validation seeds from the same generator family. The gap is therefore an upper bound on their disadvantage and must not

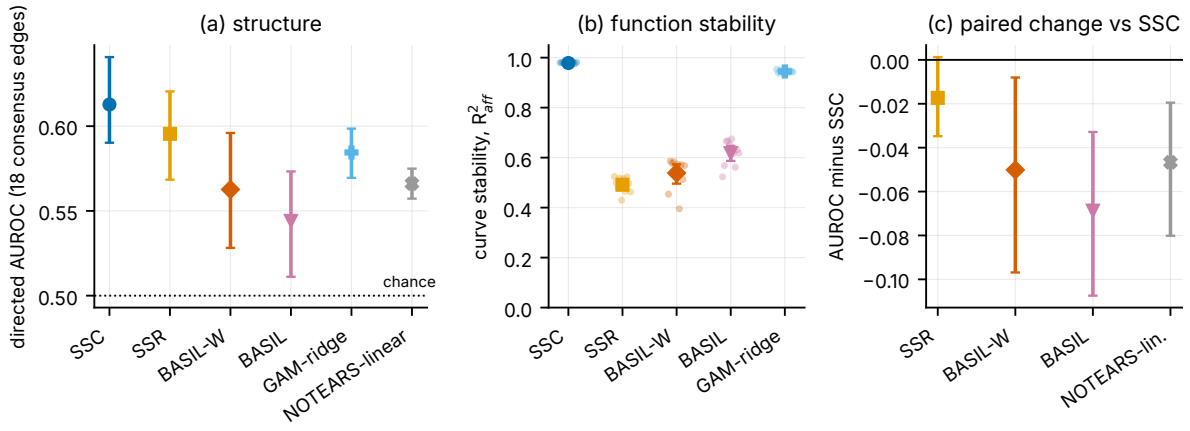


Figure 6: Sachs data ($d = 11$, 10 overlapping subsamples of 2000 cells). (a) Directed AUROC against the 18 consensus edges; (b) stability of edge curves across subsamples on a common edge set (an unselected edge counts as zero); (c) paired AUROC change relative to SPADE-scaled. Intervals are bootstrap over subsamples and optimistic.

be quoted as "published methods fail". GraN-DAG returns very sparse graphs (4.4–4.7 edges on average against 10–20 true).

Within our family at small d. BASIL-W is not better than the refit in the uniform cells (-0.023 at $d = 10$, $+0.008$ at $d = 20$, intervals include 0) and is better under Laplace noise (-0.124, $p = 0.010$); the $d = 100$ advantage over the refit is therefore not reproduced at these sizes except under heavy tails. SSA ties the refit (pooled $+0.004$). The NAM-parameterised refit is worse than the spline refit in the uniform cells (by 0.177 at $d = 10$ and 0.217 at $d = 20$, both $p \leq 0.01$) and better under Laplace noise (-0.092, $p = 0.049$); the registered prediction of "within 0.05" was wrong, but the NAM variant is untuned (one hidden layer of 16 tanh units, 400 epochs), so this says only that this untuned variant does not beat splines.

Reading and limits. The benchmark is additive with smooth one-dimensional mechanisms, which is exactly the model class of CAM-like, NAM and the spline family: the additive learners are matched to the truth and the MLP learners are not. The result is a warning that the edge-function accuracy of the SPADE family at $d \leq 20$ is not specific to it, and it is the reason the claims about BASIL-W were revised throughout this paper. CAM-like and the NAM and DCE variants are reimplementations and may differ from the originals in ways that matter; the repository NOTEARS-MLP differs from the original. Each cell has 10 datasets; best-arm choices (BASIL-W, CAM-like) were made on the evaluation data, symmetrically; p-values are uncorrected for multiplicity over 3 cells and 6 metrics, and only the TOPK NRMSE row is the registered primary. The cycle-breaking applied to cyclic own graphs was not logged. Nothing is claimed beyond $d \leq 20$, $n = 1000$, mechanism family 1, iSCM, uniform and Laplace noise and expected degree 2. Not run: KAN-based Granger methods (time-series task), SCORE and NoGAM here (structure only; they appear in the earlier tables), and, because no semi-synthetic results existed when the plan was written, a Sachs-based arm.

6.11 Real data: Sachs

On the Sachs flow-cytometry data (7466 cells, $d = 11$, natural log then z-score; 10 subsamples of 2000 cells; 18 consensus edges, Sachs et al. 2005) no repair improves structure (Table 12, Figure 6). AUROC is 0.613 for SSC, 0.595 for the refit, 0.563 for BASIL-W and 0.544 for full BASIL;

Table 12: Sachs: structure and function stability (10 subsamples).

Method	AUROC [95% CI]	AUPRC	skel. AUROC	orient.	hits in top 18	stability
SSC (SPADE-scaled)	0.613 [0.590, 0.641]	0.291	0.690	0.51	5.8	0.98
refit	0.595 [0.568, 0.621]	0.266	0.688	0.53	6.1	0.49
BASIL-W	0.563 [0.528, 0.596]	0.248	0.703	0.58	5.8	0.54
BASIL (full)	0.544 [0.511, 0.573]	0.227	0.683	0.51	5.3	0.62
GAM-ridge	0.584 [0.570, 0.599]	0.287	0.735	0.40	6.1	0.94
NOTEARS-linear	0.566 [0.557, 0.575]	0.239	0.634	0.53	5.0	–
DAGMA-linear	0.435 [0.416, 0.450]	0.168	0.688	0.31	3.0	–
var-sortnregress (z-scored)	0.503 [0.460, 0.545]	0.207	0.664	0.44	3.3	–

Sachs, $d = 11$, 10 subsamples of 2000 cells (overlapping, so intervals are optimistic), 18 consensus edges (contested; chance AUPRC 0.164, about 2.9 hits at chance). Stability: mean R^2_{aff} of the edge curves across subsamples on the 24 edges that are in SSC's top 18 in at least 3 subsamples; an unselected edge counts as a zero curve ($R^2 = 0$), which is what drives the low values for refit and BASIL-W.

orientation is about 0.5 for every method. SSC places 5.8 true edges in its top 18 on average, against about 2.9 expected at chance. Relative to SSC the refit passes the pre-registered "no loss" rule (-0.017); BASIL-W (-0.050) and full BASIL (-0.069) lose more than 0.03, and SSC beats linear NOTEARS by 0.047 AUROC. Function stability across subsamples on a common set of 24 edges is 0.98 for SSC, 0.54 for BASIL-W and 0.49 for the refit; the difference is selection instability: the refit and BASIL-W return a zero curve for about 35% of (edge, subsample) cells, SSC never does, and when an edge is kept in both subsamples all methods are stable (R^2_{aff} 0.96–0.98). By the pre-registered rule the refit and BASIL-W are therefore "less stable" than SSC. The amplitude of the repaired curves relative to SSC is 1.60 (refit) and 1.41 (BASIL-W), i.e. the shrinkage predicted by the simulations is present but smaller than the factor of about 2.3 that the synthetic slope would imply, and there is no ground truth to say which amplitude is right. Sign sanity checks against three textbook relations were uninformative or unsupportive. This is a negative result for using the repairs as-is on real data. High stability of SSC reflects in part that it is a smooth always-on fit; stability is not correctness and not evidence of causal meaning (only two of the six top SSC edges are consensus edges in the consensus direction).

6.12 Negative results

Table 13 collects the attempts that did not work. Three deserve a sentence each.

SPIRE. The first idea of the project was a single objective: a group-MCP penalty, which has no shrinkage on strong edges by construction, plus a residual-independence (HSIC) term to exploit the ANM identifiability signal. At $d = 10$ with 30 test datasets it lost 0.021 AUROC against SPADE and reached a slope of 0.75 (kill line 0.80) while the old adaptive heuristic reached 0.93 without an AUROC loss; the HSIC term added at most 0.05 orientation and was not significant on validation. The suspected reason is that exact zeros from the proximal step discard ranking information that SPADE's dense soft fit keeps; this is a hypothesis, and a convergence-budget test did not change the result. The independence-plus-adaptive-group-sparsity idea is also pursued by Yao et al. (2026).

BASIL-S and full BASIL. Full BASIL took about 11 minutes per dataset at $d = 50$. The screened variant BASIL-S was 57 times faster at $d = 50$ and passed its faithfulness criterion, but with random starts its two chains disagreed by hundreds to thousands of nats at $d \geq 100$, orientation collapsed at $d = 200$ (0.830 against 0.942 for SPADE-scaled), and at $d = 500$ (one dataset) orientation was 0.714 and the slope 0.51. An amended version (greedy initialisation, $3\times$ longer chains, chain selection) was developed on validation seeds and recovered orientation but not

Table 13: Negative and failed results kept in the paper.

Attempt	Setting	Outcome	Verdict
SPIRE (group-MCP + HSIC term)	$d = 10$, $n = 1000$, 30 datasets	AUROC -0.021 $[-0.031, -0.010]$ vs SPADE; slope 0.75 (kill line 0.80); HSIC term adds $+0.05$ orientation at best, not significant on validation	H1 fails, H2 inconclusive
BASIL-S cold start	$d = 100$–500, 5 datasets (1 at $d = 500$)	chain end-scores differ by hundreds of nats; AUROC -0.02 to -0.03 vs SPADE-scaled; orientation 0.83 vs 0.94 at $d = 200$ (original); $d = 500$: orientation 0.71, slope 0.51	does not scale as specified
BASIL-S, cold start (ablation)	$d = 100$, 8 datasets	NRMSE 0.324 vs 0.177, orientation 0.809 vs 0.943 (8/8 worse)	warm start essential
Full BASIL	$d = 50$	about 11 min per dataset; no better than BASIL-W on Sachs (AUROC 0.544)	not scalable
SCORE / NoGAM	$d = 10$, 10 datasets	AUROC 0.858 / 0.895 vs 0.980 (SPADE), 5–10× slower	baseline only
Pre-registered BASIL-W do-effects	$d \geq 100$, 4 of 45 datasets	indirect do-effect NRMSE up to 185 (coefficients up to 1.4×10^7); fixed post hoc by a ridge map	instability; fixed, 0/75 on fresh seeds

Numbers from research/spire (README_results, README_basil, README_scale, README_do), research/ablation and research/real_data.

AUROC (still 0.02–0.03 below) and was no faster than SPADE-scaled at $d = 200$. SPADE-scaled itself already scales (AUROC 0.997–0.999 at $d = 100$, 200, 500; 0.7–9 minutes per dataset), so scalability is not a differentiator for BASIL. The solution that works (BASIL-W) starts from SPADE-scaled's order, which makes it a refinement of the learner, not an independent estimator.

Numerical instability of the pre-registered do-effects. Four of 45 pre-registered BASIL-W datasets gave indirect do-effect NRMSE far above 1 (up to 185). The cause, diagnosed on one case, was coefficients up to 1.4×10^7 in the whitened-to-spline coefficient map along weakly excited spline directions, which makes propagated interventions leave the data support. A ridge-stabilised map removed it (0 of 75 datasets above NRMSE 2 on fresh seeds; Section 6.3). The stabilisation was chosen after seeing test results, with a single value of its constant, and confirmed on fresh seeds.

Other. SCORE and NoGAM are 0.07–0.11 AUROC below SSC at $d = 10$ and 5–10 times slower; the GAM-ridge baseline without DAG constraint has similar skeleton AUROC to SPADE but orientation 0.29 lower (0.65 against 0.95), so the DAG constraint matters for orientation, not for the skeleton.

7 Discussion and limitations

What the evidence supports. (i) Structure-accurate sparse spline learners can return edge functions with less than half of the true amplitude at d up to 500, across noise families, sample sizes and two mechanism families, and this carries over to interventional effects. (ii) The cheap fixes work: the adaptive group lasso removes most of the error without changing the selected structure, and the refit removes about two thirds of it at a structural cost. (iii) A Bayesian model-averaged estimator warm-started from the learner's order removes somewhat more, is

more robust to heavy tails and heteroscedasticity than the refit, and supplies calibrated edge probabilities, at the price of AUROC and a more complex pipeline. We do not recommend BASIL-W over the adaptive group lasso as a default, and we do not claim that BASIL-W is the best learner. (iv) On independent DAGs BASIL-W beats the refit by only 0.03–0.04 NRMSE under uniform and heteroscedastic noise (the refit wins 4/20 and 6/20 datasets), and at $d \leq 20$ an additive spline learner from the literature (our own Python reimplementation of CAM) has lower or equal edge-function error than BASIL-W (Laplace $d = 10$: 0.151 against 0.281, $p = 0.002$; pooled over 30 non-independent datasets 0.161 against 0.212, $p = 0.013$), while the untuned neural learners are far behind. The contribution that stands is the diagnosis of amplitude shrinkage in sparse spline DAG learners, the evidence that standard fixes remove most of it, and the check at scale (d up to 500), where no competitor was tested; it is not a claim of superiority of a particular estimator. (v) On a semi-synthetic benchmark from Sachs the BASIL-W/SSC error ratio is 0.40–0.47 at $n = 2000$ but 0.78–0.81 at $n = 500$, a plain ridge GAM ties or beats BASIL-W at $d = 11$ in three of four cells, band coverage is 0.39–0.63, and the real-data pattern is not reproduced.

Limitations (complete list).

1. Simulation is the only place where function ground truth exists; the real-data evidence is negative or ambiguous, and the semi-synthetic benchmark (Section 6.9) is a fitted model of contested data (17 true edges after removing one cycle edge by a pre-fixed rule), uses the spline family of the learners, and agrees with the simulation story only partly.
2. The external comparison (Section 6.10) covers $d \leq 20$ only, ten datasets per cell, an additive benchmark that matches the additive learners, an own Python reimplementation of CAM (not the original R code), untuned neural learners run at default or repository settings, and uncorrected tests; it does not show whether BASIL-W or CAM-like is better at $d \geq 100$.
3. Ten datasets per cell in the first replication, with cells C1–C6 sharing the same 10 DAGs, so pooled p-values overstate the evidence; the independent-DAG replication has 20 datasets per cell and three noise families only.
4. A single generator family (iSCM with Erdős–Rényi graphs of expected degree 2, two mechanism families, four noise types); no other graph densities, scale-free graphs, non-additive mechanisms, latent confounding or measurement error; mechanism functions are smooth and low-order.
5. $d = 500$ rests on two datasets and, in the first grid, on the BASIL-W version before stabilisation; the adaptive arm at $d = 500$ is also two datasets (Section 6.6).
6. The ridge stabilisation of BASIL-W's do-effects was chosen after seeing test results with one value of its constant; it was confirmed on fresh seeds (0 of 75 explosions) but never tuned.
7. The adaptive group lasso used a single untuned setting of γ and ϵ in the main grid, and was run on 3–10 datasets per cell; the sensitivity study is small (five datasets, no significance claims); its seeds were reused from earlier runs, so it is a sensitivity analysis and cannot select a setting, and two grid settings that look better were not adopted.
8. The novelty of the repairs is limited (Section 2.1); the literature check relied on web search and abstract-level reading. The concurrent ICML 2026 paper (Yao et al., 2026) was read via its abstract only; Kuskova et al. (2026) likewise.
9. Credible bands are valid only on average: the calibration constant κ^* was chosen on five validation datasets and was somewhat lucky; found edges are over-covered, missed edges remain uncovered; simultaneous coverage is low.

10. BASIL-W inherits its candidate set and its starting order from SPADE-scaled; if SPADE-scaled's order is poor, BASIL-W's behaviour is untested. Chains are not provably converged: four chains do not agree better than two.

11. BASIL-W's AUROC is below SPADE-scaled's in essentially every dataset (0.006–0.032 on average) and two cells miss the pre-registered 0.03 target by 0.001–0.002.

12. The Sachs consensus graph is contested, the data pool interventional conditions that the methods do not use, $d = 11$ with 18 edges makes 0.05 AUPRC about one edge, the 10 subsamples overlap by about 27% (intervals optimistic), and there is no ground truth for the functions. The three sign-sanity relations were taken from memory of the standard cascade and not verified against the original text.

13. Order MCMC has a prior bias towards graphs compatible with many orders (Viinikka et al., 2020), which we do not correct; the Gaussian likelihood and $k_{\max} = 3$ limit BASIL-W; model errors, not sampler errors, dominate in some datasets (the true order scored 25–46 nats below the MCMC order on an early validation dataset).

14. The refit multiplier $k = 1.5$ was tuned on three validation datasets and exploits that the generating graphs have mean degree 2; a user may not know the density. Refit and adaptive baselines were given the screened candidates and $\lambda = 0.2/d$ of SPADE-scaled, whose constants were fixed on an earlier generator.

15. In the independent-DAG replication BASIL-W's advantage over the refit is 0.039 and 0.027 NRMSE in the uniform and heteroscedastic cells (the refit wins 4/20 and 6/20 datasets), direct do-effects of BASIL-W and SSA are tied under uniform noise (Holm $p = 0.07$), band coverage is 0.76–0.79 against the nominal 0.90, and γ and ϵ of SSA remain untuned, the largest open threat to the BASIL-W-over-SSA claim.

16. At $d = 500$ the adaptive arm is two datasets, and SSA's do-effects there are worse than the refit's, contrary to the registered prediction.

17. The BASIL-W ablations are on 8 datasets without significance tests, and three registered predictions were wrong.

18. Cost ratios were measured on a shared, loaded machine and may not transfer; the 4-core CPU timings exclude metric code.

19. Some of the independent-DAG replication's uniform datasets (seeds 3000–3012) coincide with the seeds used for the band-calibration and ablation work at $d = 100$ (validation 3000–3004, test 3005–3012); no BASIL-W setting was changed by that work, but those datasets are not untouched (Section 6.5).

20. Do-effect evaluation covers single-node hard interventions on a 15-point grid and at most 20 direct and 20 indirect pairs per dataset; pairs with tiny true effects are skipped (about 10% of pairs, identical across methods).

Risks of misreading. The largest NRMSE reductions are relative to a learner (SPADE-scaled with $\lambda = 0.2/d$) that is deliberately sparse; a less sparse baseline would shrink less. Our result is about the shape bias of sparse group-penalised fits, not about any specific software.

8 Conclusion

Finding the edges is not enough: on iSCM data a screened group-lasso spline learner has near-perfect structure but returns curves with about 44% of the true amplitude, and its interventional effects are wrong even for direct parents. Standard repairs work. The adaptive group lasso gets

most of the way without hurting structure; a refit gets two thirds of the way at a structural cost and breaks down under heavy-tailed noise; a warm-started Bayesian model average goes a little further (by 0.03–0.04 NRMSE over the refit under uniform and heteroscedastic noise, more under heavy tails), adds calibrated edge probabilities, but costs AUROC, has under-covering bands and, on the one real dataset we could test, is less stable than the unrepaired learner. It is not the best learner overall: at $d \leq 20$ an additive spline baseline (our own Python reimplementation of CAM) matches or beats it, and a ridge GAM does so in some settings of a semi-synthetic benchmark at $d = 11$. Reporting edge-function error next to structure metrics costs little and changes the ranking of methods. The evidence is simulation-based with one generator family and small cell sizes; the semi-synthetic benchmark agrees with the simulations only partly, and the real-data evidence is negative.

Reproducibility statement

Code and data. All code, plans, raw per-dataset CSVs and analysis scripts are in the git repository, under `research/` (repository location and release commit: [**to be inserted by the authors at release**]). Datasets are generated from seeds by `bench.gen_full`; the Sachs data are in `data/sachs/` (7466 cells, 11 variables, consensus graph with 18 edges). **Regenerating this paper.** From `research/paper/ms/`: `python3.13 make_example_curves.py` (about two minutes, one core), `python3.13 make_figures.py`, `python3.13 make_tables.py`, then `pdflatex main; bibtex main; pdflatex main; pdflatex main`. **Re-running the experiments.** Each experiment directory has a job list and runner: for example `research/replication/run_all.sh` followed by `python3.13 aggregate.py`; the replication of 75 datasets takes several hours on 2 jobs of a 4-core CPU. **Seeds.** Validation 0–9 and 3000–3004; earlier tests 1000–1004 and 1000–1029 ($d = 10$); confirmatory 2000–2009 (C8: 2000–2004); ablation tests 3005–3012; Sachs subsamples seeds 0–9 (`RandomState(r).choice(7466, 2000)`); method seed 0 everywhere. One bit-identical re-run check on two datasets confirmed that the replication code reproduces the earlier numbers. **Hardware and software.** CPU only, a shared 4-core machine (no GPU), one thread per job, at most two parallel jobs; Python 3.13, PyTorch, NumPy, scikit-learn, SciPy. Timings are noisy.

Declarations

Author contributions. [**To be completed by the authors, e.g. following the CRediT taxonomy.**]

Competing interests. [**To be completed by the authors.**]

Funding. [**To be completed by the authors.**]

Data availability. [**To be completed by the authors. Synthetic data are generated by the released code from stated seeds; the Sachs data are a public benchmark (source and licence to be stated).**]

Use of AI assistance. [**Placeholder to be reviewed and adapted by the authors to the target venue's policy. Facts: an AI assistant (Claude, Anthropic) was used to write code, run experiments, analyse results and draft text, including the literature search and verification notes and this manuscript. The authors are responsible for the content and must verify all numbers and references.**]

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A Pre-registered hypotheses and verdicts (confirmatory replication)

Table 14: Verdicts against `research/replication/PLAN.md` (thresholds copied from the plan).

Hypothesis	Verdict	Evidence
H1: NRMSE ratio BASIL-W/SSC ≤ 0.70, every cell; kill ≥ 0.90	confirmed (8/8)	ratios 0.22–0.38, upper CI ≤ 0.40
H2: BASIL-W $\leq$ refit $+0.03$ in ≥ 5 of 7 cells	confirmed (7/7)	lower in all 7 means; Holm-significant in 4; tie in C1, C5, C6
H3: do-effect ratio ≤ 0.70 (direct, indirect), no indirect error > 2	confirmed	direct 0.19–0.30, indirect 0.24–0.52; 0 of 75 above 2
H4: AUROC BASIL-W $-$ SSC ≥ -0.03 in every cell; kill < -0.05	not confirmed; kill not reached	C2 -0.032, C6 -0.031 miss; 6 of 8 pass
H5: coverage ≥ 0.80 and ECE ≤ 0.10	ECE yes; coverage not confirmed	coverage 0.74–0.81, below 0.80 in 7 of 8 cells; ECE ≤ 0.0024

B Deviations from the plans

- *BASIL-S.* Chains were amended after test diagnostics (greedy start, longer chains, chain selection), tried on three validation seeds, then both configurations were run on test seeds.
- *Do-effects.* The ridge-stabilised coefficient map was chosen after seeing test results and re-run only on the cells that had exploded; the replication is its first clean test.
- *Replication.* Timing columns changed (cost excludes metric code); results of two datasets re-run and bit-identical to the earlier files.
- *Sachs.* Same subsamples and seed as earlier non-registered runs (disclosed in the plan); one variance-sorting baseline on the pre-z-score scale was added outside the plan; a multi-threading problem forced a restart pinned to one core.
- *SPIRE.* Several solver details deviate from the formulation document (reduced basis size, restricted HSIC gradient, step schedule, tie-breaking, fixed σ); an early validation run with a larger penalty constant was overwritten and is not reported.
- *Ablations.* Validation datasets turned out to be easier than test datasets (fewer missed edges), which affects the calibration constant.